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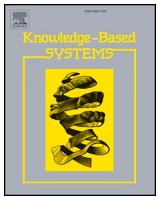

Multiscale-attention masked autoencoder for missing data imputation of wind turbines

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ABSTRACT

High-quality data is essential for effective operation and maintenance of wind farms. However, data missing is a persistent issue in the supervisory control and data acquisition (SCADA) system, which seriously affects the data quality. To tackle the two limitations of current missing data imputation methods: the gap between training tasks and imputation tasks, and the inadequate extraction of correlations within SCADA data, this work proposes a data-driven framework named multiscale-attention masked autoencoder (MAMAE) for missing data imputation of wind turbines. The MAMAE employs masked autoencoding as a self-supervised training method, bridging the gap between the training and imputing task. Additionally, considering the importance of correlations in imputation for the SCADA data, a multiscale attention architecture built upon transformer is employed. Comprising four transformer stages, each applying attention mechanisms at distinct scales, the multiscale attention efficiently extracts feature, turbine, and temporal correlations. To ameliorate the problem of large computation cost caused by increased sequence length in different scales, localized attention is implemented in shifted windows, reducing the computational complexity from quadratic to a linear relationship with the sequence length. Furthermore, a turbine correlation-based feature combination method is proposed to coordinate with the multiscale attention and introduce turbine correlations into the imputation process. Experiments were conducted on a SCADA dataset collected in a real-world wind farm. The results show that the proposed method achieves higher accuracy than existing methods in most cases (especially in the cases with band missing and feature missing) and the ablation experiments verify the effectiveness of each proposed modification in improving accuracy or efficiency.

1. Introduction

Over the past few decades, wind power has experienced rapid development as one of the most mature renewable energy sources with significant large-scale commercialization prospects. Despite this growth, the harsh installed environment and unstable power generation present challenges to wind power deployment. To address these issues, effective operations and maintenance (O&M) have gained prominence, encompassing wind power prediction [1,2], condition monitoring [3,4], fault diagnosis [5-8], and other applications. Supervisory control and data acquisition (SCADA) systems are extensively used to provide operating data for wind turbines in these applications. However, challenges arise due to communication failures, sensor malfunctions, and the need for data cleaning of abnormal data [9,10]. These issues result in data incompleteness, which negatively impacts data quality and may cause kinds of problems in control strategy, decision making, maintenance planning, etc. [11]. To ensure high-quality O&M, imputation of missing data for wind turbines is of significant importance.

Existing imputation methods can be primarily categorized into four types: probability-based methods [12,13], interpolation-based methods [14], similarity-based methods [14-17], and regression-based methods [17-27]. Probability-based approaches, represented by the expectation-maximization (EM), rely on artificially defined probability distributions [23]. Interpolation-based methods, such as linear and cubic interpolation, employ polynomials to model the measurements using time dependency and fill the missing points with the combinations of the neighbors, rendering them fast and data-free. These methods exhibit considerable accuracy when missing values are sparse, but the performance deteriorates significantly in the presence of numerous missing values [28]. The limitation stems from the fact that

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interpolation-based methods account for only temporal correlations, where feature correlations are mostly neglected. Similarity-based methods (including k-nearest neighbors (KNN), K-means, and mean value imputation) calculate similarities between missing and complete samples based on feature correlations and then use the mean value of highly similar complete samples to impute missing values. However, similar to interpolation-based methods, neglecting temporal correlations results in information loss and accuracy decrease [24]. Regression-based methods, particularly shallow and deep learning methods, can systematically consider both feature correlation and temporal correlation once inputs and outputs are appropriately designed. Benefitted from the outstanding capacity in modeling highly nonlinear problems, deep learning approaches excel in extracting complicated correlation features, thereby achieving superior imputation accuracy. Moreover, the end-to-end modeling capability allows for convenient pipeline design compared to traditional regression methods [17-19].

Deep learning methods for missing data imputation can be summarized into three categories by training strategy: autoregression [20-22], autoencoding [23,24] and adversarial training [25-27]. Generally, autoregression captures the time-dependent relationships and forecasts time steps relying on previous ones. For missing data imputation, the framework is widely built with the recurrent neural network (RNN) and its derivatives such as long short-term memory (LSTM) and gate recurrent unit (GRU), utilizing complete or imputed time steps to predict missing time steps. Autoencoding is used for missing data imputation as of its reconstruction ability that the autoencoding framework could corrupt the input in a proper manner (mostly compressing the input into a low dimensional embedding) and reconstruct the embedding into the original input. The most common autoencoding model is the autoencoder (AE), which compresses the high-dimensional input into a lower-dimensional hidden state using an encoder and then reconstructs the hidden state using a decoder. The encoder and decoder can be multi-layer perception [23], convolutional neural networks [24] or other neural networks. Adversarial training, a technique foundational to generative adversarial networks (GAN), is leveraged in GAN-based imputation methods. These methods entail the adversarial training of a generator and a discriminator, where the generator is trained to produce data that conforms to the distribution of the complete data using random vectors. The generated data is then employed to estimate and fill in missing values within the original dataset. However, all these techniques employed in missing data imputation have limitations. Autoregression leverages RNN's strength in temporal correlations and imputes missing values through time series prediction, which lacks a global receptive field [29] and results in error compounding [30]. Autoencoder inevitably experience information loss due to the bottleneck structure. Meanwhile, GAN-based methods focus on the data generation according to the learned distribution and encounter convergence difficulty and mode collapse [31]. More fundamentally, the training task for these methods are not intuitive, which exhibit discrepancies with the imputation task. These inconsistencies between the training and imputation processes limit the models' generalization performance.

With the rapid advancement of pre-training techniques in natural language processing (NLP) and computer vision (CV), masked autoencoding has achieved great success and emerged as one of the most prevalent pre-training methods. Unlike the compression process of AE, masked autoencoding masks some tokens in the input and then use neural networks to restore masked content. Represented by bidirectional encoder representation from transformers (BERT) [32] in NLP and masked autoencoders (MAE) [33] in CV, the masked autoencoding has exhibited a strong capacity to reveal patterns within data. The masking and reconstructing task in the masked autoencoding is akin to missing data imputation, which leads to the intuition that the masked autoencoding could be a more suitable training strategy for missing data imputation compared to traditional techniques. In this context, the primary motivation of this work is to enhance the data imputation performance by substituting autoencoding with masked autoencoding as the training task.

Masked autoencoding models typically builds with the transformer architecture [34], which has achieved great success in various fields, such as generative pre-trained transformer (GPT) [35-37] in natural language processing, vision transformer (ViT) [38] in computer vision, DALL-E [39,40] in text-to-image generation, and industrial applications [41,42]. The key component of the transformer is the attention mechanism [43,44], which extracts correlations between tokens in a sequence. It can be effectively employed to extract various correlations within wind turbine data, addressing the limitations of existing methods that struggle to fully utilize crucial multiscale correlations without specifically designed correlation extraction algorithms. The comprehensive correlation ability is essential for imputation, as the dominant imputation clues may involve different types of correlations in various missing data scenarios. Most existing methods rely on fully connected layers to extract correlations, while some sequence-based models, including the traditional transformer, only introduce advanced algorithms for temporal correlation extraction. Therefore, a multiscale attention architecture is proposed, which explicitly applies the attention mechanism to tokens of varying scales, aiming to enhance the model's comprehensive correlation extraction ability.

By incorporating masked autoencoding and the multiscale attention, this work proposes a novel framework for missing data imputation of wind turbines, noted as the multiscale-attention masked autoencoder (MAMAE). Except for the masked autoencoding and the multiscale attention, the MAMAE also introduces a preprocessing technique called feature combination to further incorporate correlations among wind turbines into the imputation process and coordinate with the multiscale attention. The contributions of the proposed imputation framework (MAMAE) include:

1) A novel multiscale-attention masked autoencoder (MAMAE) for missing data imputation of wind turbines is proposed. The MAMAE incorporate the masked autoencoding as the training task and the shifted-window-based multiscale attention architecture as the correlation extractor, bridging the gap between training and imputation procedures and leveraging multiscale correlations effectively with relatively low additional computational cost.
2) Based on the correlation coefficients among distinct wind turbines within the wind farm, a feature combination technique is proposed to introduce turbine correlations into the imputation process. This approach is aimed at improving the imputation accuracy for large-scale and complete data missing in individual wind turbines.
3) Experiments are conducted on a real-world SCADA dataset, including accuracy assessment and ablation studies. The experimental results show that the MAMAE outperforms the comparison approaches across most tested missing situations, demonstrating higher accuracy, adaptability and generalization ability compared to traditional imputation methods, and the ablation studies verify the effectiveness of each proposed modification in improving accuracy and efficiency.

The remaining sections are outlined as follows. Section 2 presents the problem formulation of wind data imputation; Section 3 elaborates on the MAMAE framework and relevant methods; Section 4 describes the implementation of experiments based on a real-world SCADA dataset, along with the presentation of results. Finally, the conclusions are given in Section 5.

2. Problem formulation

2.1. Data missing

Considering a multivariate time series sample of SCADA data

collected in wind turbines represented as $\boldsymbol{X} = (x_{ij}) \in \mathbb{R}^{T\times D}$, where T denotes the number of time steps and D represents the number of features. Given that missing values are likely present, a missing matrix $\boldsymbol{M} = (m_{ij}) \in \mathbb{R}^{T\times D}$, of the same size as $\boldsymbol{X}$, is introduced to indicate the absence of data. In this matrix, $m_{ij} = 0$ indicates that x_{ij} has been observed, while $m_{ij} = 1$ signifies a missing value. This is quantified by Eq. (1):

$$m_{ij} = \begin{cases} 0 & \text{if } x_{ij} \text{ is observed} \\ 1 & \text{if } x_{ij} \text{ is missing} \end{cases} \tag{1}$$

A sample is regarded as complete if $m_{ij} = 0,\ \forall i \le T,\ j \le D$ or as incomplete otherwise. If SCADA data has been made into a dataset of time series, the samples in the dataset can be grouped into a complete sample set where all samples are complete and an incomplete sample set which consists of incomplete samples.

For a cluster of wind turbines, the features:

$$\boldsymbol{\Lambda} = \{\lambda_{mk}\},\ m \in \{1, 2, \cdots,\ M\},\ k \in \{1, 2, \cdots,\ K\} \tag{2)}$$

are naturally grouped into a number of sets $\boldsymbol{\Lambda_k}$,

$$\boldsymbol{\Lambda_k} = \{\lambda_{mk}\},\ m \in \{1, 2, \cdots,\ M\} \tag{3}$$

where M is the number of measurements of each wind turbine, K is the number of wind turbines, and $D = M \times K$.

In the operating data collected in the wind farm, the data is also categorized into sets according to the measurements of each wind turbine. The data missing may occur in sets as of physical constraints of the wind turbines.

2.2. Types and ratios of missing data in wind turbines

Factors such as communication failures, sensor malfunctions, and data cleaning results in various missing types that significantly impact model evaluation and the validity of experiments. This work examines three distinct missing types for experimental purposes:

(i) Random missing. Random missing refers to the random positioning of missing values and is a common benchmark in general imputation studies. Although not frequently observed in SCADA data, it serves as a fundamental missing type for evaluating model performance.

(ii) Band missing. Band missing denotes a scenario where all features of a wind turbine are absent for a specific time period, presenting as bands in missing matrix.

(iii) (Feature missing. Feature missing refers to a situation where one feature is missing for all wind turbines temporarily. In contrast to random missing, band and feature missing are both temporally continuous.

Type (ii) and Type (iii) provide a more realistic assessment of the model's practical performance as they occur more frequently in SCADA data. Fig. 1 displays missing matrix examples for the three missing types. Notably, band missing is also a special case of data loss in correlated attributes (DLCA) in Ref. [23] and continuous missing in Ref. [24].

The scale of missing data significantly impacts the accuracy and applicability of models. Consequently, this work employs the missing ratio as a quantitative descriptor. The missing ratio (MR) is defined as follows:

$$MR = \frac{\sum_{ij} m_{ij}}{TD^{\mathrm{m}}} \tag{4}$$

In accordance with the respective missing data types, D^{m}, denoting the number of features with likely missing values, equals D for random missing, M for band missing, and K for feature missing.

3. Methodology

3.1. Overall structure

The proposed imputation framework consists of a preprocessing stage and an imputation stage using the MAMAE model. Both stages implement techniques to enhance imputation performance by integrating advanced training tasks or capitalizing on multiscale correlations. The preprocessing stage is employed to prepare time series samples for the MAMAE model. Importantly, it contains the feature combination as the core component, which concatenate the features of highly-correlated turbines. The efficacy of the feature combination can be anticipated due to the fact that nearby wind turbines often experience comparable environmental conditions—such as terrain, wind patterns, and temperature—and demonstrate analogous power output traits when operating in the same mode. Consequently, their data are inherently informative about one another. By concatenating the features of these highly-correlated turbines, the combination features contain supplementary information from related turbines, thus enabling the utilization of turbine correlations. This method is especially beneficial in cases with band missing data, where the target turbines exhibit diminished temporal and feature correlations.

The MAMAE model's overall structure is illustrated in Fig. 2. The model is designed to employ the masked autoencoding as the training

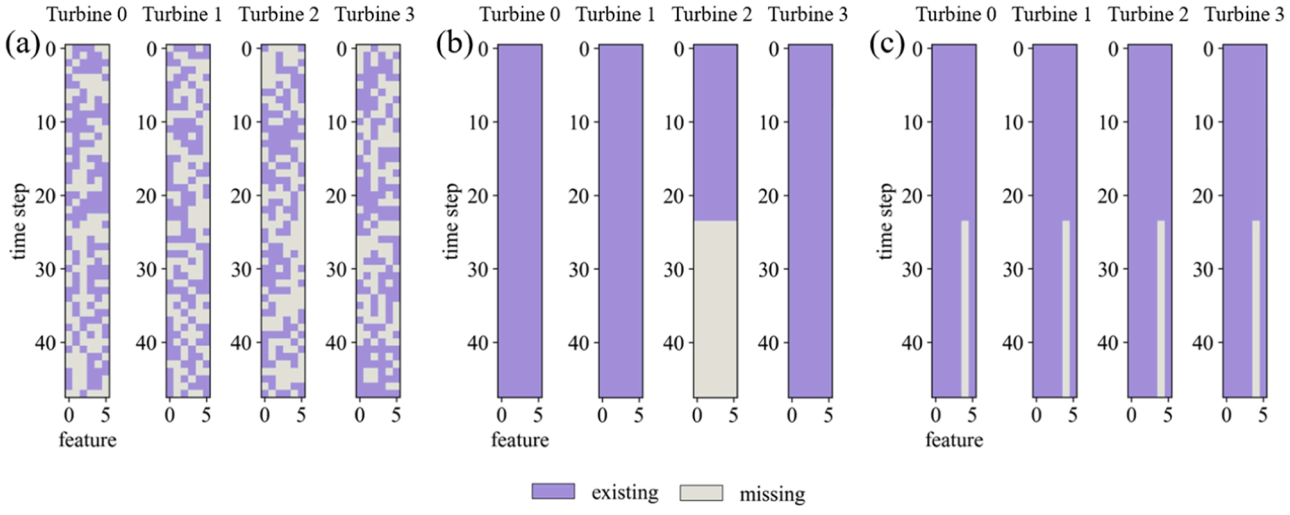

Fig. 1. Missing matrix examples of different missing types. (a) Random missing. (b) Band missing. (c) feature missing.

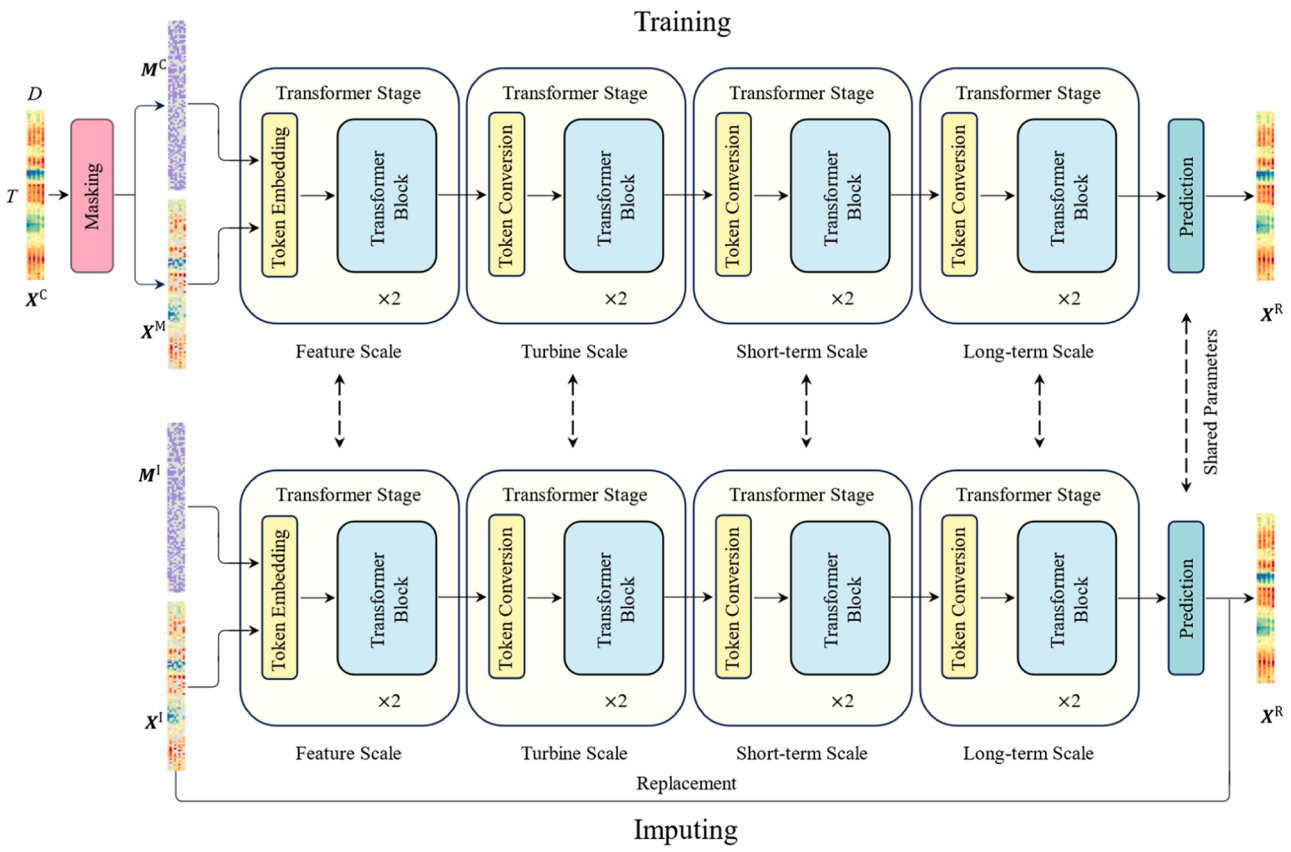

Fig. 2. Overall structure of the MAMAE model.

task and a shifted-window-based multiscale attention architecture as the correlation extractor. The masked autoencoding aims to bridge the gap between the training and imputing process. During the training process, a predefined percentage of input data is artificially masked by a masking module using a predefined pattern before being fed into the model, which is then tasked with predicting the masked values. The training task closely mirrors the imputation task, compelling the model to investigate potential patterns of missing data and to uncover the imputation capabilities. The multiscale attention architecture is designed to more effectively extract multiscale correlations by explicitly applying attention mechanism on feature, turbine, short-term, and long-term scale. This is realized by four stages of modified transformer, each comprising a token conversion module designed to generate sequences of specialized physical representations and multiple transformer blocks that are intended to apply attention mechanism specifically to the given scale. To reduce the additional computation cost caused by increased sequence length in specific stages, shifted window-based multi-head self-attention is employed to replace the original attention mechanism in transformer blocks.

The overall process of the proposed imputation method is as follows. During the preprocessing stage, raw SCADA data are initially processed through the feature combination technique to create combined features that reflect the high correlations among turbines. Next, a sliding window approach is applied to these combination features to construct a time series dataset. This dataset is finally separated into complete samples ($\boldsymbol{X}^{\mathrm{C}}$, the samples without missing values) and incomplete samples ($\boldsymbol{X}^{\mathrm{I}}$, the samples with missing values). The complete samples are used to train the MAMAE model, which is subsequently employed to impute the missing values within the incomplete samples. It should be noted that the behavior of the MAMAE model differs between the training and imputing phases. During training, input data $\boldsymbol{X}^{\mathrm{C}}$ is partially masked in a specific pattern, denoted as $\boldsymbol{X}^{\mathrm{M}}$, generating a masking matrix $\boldsymbol{M}^{\mathrm{C}}$ that shares the same definition as the missing matrix. The four consecutive transformer stages, subsequently applies attention mechanisms to $\boldsymbol{X}^{\mathrm{M}}$ and $\boldsymbol{M}^{\mathrm{C}}$ in four distinct scales: feature, turbine, short-term, and long-term. This process extracts multiscale correlations and produces a latent representation. Finally, a prediction module maps the latent representation to a reconstructed input $\boldsymbol{X}^{\mathrm{R}}$, which is of the same shape as the original input. Loss functions and metrics, such as mean squared error (MSE), can be calculated using $\boldsymbol{X}^{\mathrm{C}}$ and $\boldsymbol{X}^{\mathrm{R}}$. Upon acquiring the correlation extraction and imputation abilities by training, the model can be utilized for imputing incomplete data. First, the missing matrix $\boldsymbol{M}^{\mathrm{I}}$ of incomplete samples must be obtained by examining the positions of missing values. Next, the incomplete samples and missing matrix are processed directly by the transformer stages and prediction module without artificial masking, yielding reconstructed outputs. As neural networks cannot guarantee constant existing values, the original existing values in $\boldsymbol{X}^{\mathrm{I}}$ are employed to replace the corresponding values in the reconstructed output. The overall process is summarized in Algorithm 1. By maintaining the shape of input and output consistent, the MAMAE model emerges as a straightforward end-to-end imputation method, enabling imputation across diverse missing data types without the need for specialized design.

3.2. Feature combination

Feature combination in preprocessing stage is crucial for introducing and leveraging correlations between turbines. Initially, correlation coefficients between distinct turbines should be computed using an artificially chosen feature. Suppose the missing values of the $\hat{n}_{th}$ turbine are to be imputed, and the m_{th} feature is selected for calculating the correlation coefficient. The time series of the $\hat{n}_{th}$ turbine's m_{th} feature is defined as $\boldsymbol{Y}_m^{\hat{n}}$:

Algorithm 1
Multiscale-attention Masked Autoencoder for Missing Data Imputation.

Input: $\boldsymbol{Y}$-the raw SCADA data matrix, N_{wt}-the number of wind turbines, $\mathcal{L}(\cdot)$-loss fuction, N_{epoch}-the number of epochs.
Output: $\boldsymbol{X}^{\mathrm{R}}$-imputed incomplete samples.
// Part 1: Preprocessing
for $\hat{n} = 1, \ldots, N_{wt}$ **do**
  Generate $\boldsymbol{X}^{\hat{n}}$ from $\boldsymbol{Y}$ using Eq. (7-9); // Feature combination.
end for
Generate time series samples $\boldsymbol{X}^{*}$ from $\{\boldsymbol{X}^{1}, \boldsymbol{X}^{2}, \ldots, \boldsymbol{X}^{N_{wt}}\}$ using a sliding window.
Divide samples $\boldsymbol{X}^{*}$ into complete samples $\boldsymbol{X}^{\mathrm{C}}$ and incomplete samples $\boldsymbol{X}^{\mathrm{I}}$;
// Part 2: Train MAMAE model
for $epoch = 1, \ldots, N_{epoch}$ **do**
  Generate $\boldsymbol{M}^{\mathrm{C}}$ according to masking strategy;
  Map $\boldsymbol{X}^{\mathrm{C}} \rightarrow \boldsymbol{X}^{\mathrm{M}}$ using Eq. (11); // Masking.
  for $stage = 1, \ldots, 4$ **do**
    if $stage = 1$ **then**
      Map $\boldsymbol{X}^{\mathrm{M}}, \boldsymbol{M}^{\mathrm{C}} \rightarrow \boldsymbol{x}^{\mathrm{E}}$ using Eq. (12); // Token embedding.
    else
      Map $\boldsymbol{x} \rightarrow \boldsymbol{x}^{\mathrm{E}}$ using Eq. (13); // Token conversion.
    end if
    Map $\boldsymbol{x}^{\mathrm{E}} \rightarrow \boldsymbol{x}$ using Eq. (14); // Transformer block with W-MSA.
    Map $\boldsymbol{x} \rightarrow \boldsymbol{x}$ using Eq. (14); // Transformer block with SW-MSA.
  end for
  Map $\boldsymbol{x} \rightarrow \boldsymbol{x}^{\mathrm{O}}$ using Eq. (18); // Reconstruct samples using latent representation.
  $\boldsymbol{X}^{\boldsymbol{R}} = \boldsymbol{x}^{\mathrm{O}}$; // Reconstructed samples are equal to $\boldsymbol{x}^{\mathrm{O}}$ while training.
  $l = \mathcal{L}(\boldsymbol{X}^{\mathrm{C}}, \boldsymbol{X}^{\mathrm{R}})$; // Calculate loss.
  Update model parameters.
end for
// Part 3: Impute incomplete samples using MAMAE model.
Generate missing matrix $\boldsymbol{M}^{\mathrm{I}}$ from $\boldsymbol{X}^{\mathrm{I}}$ using Eq. (1) and replace missing values with 0;
for $stage = 1, \ldots, 4$ **do**
  if $stage = 1$ **then**
    Map $\boldsymbol{X}^{\mathrm{I}}, \boldsymbol{M}^{\mathrm{I}} \rightarrow \boldsymbol{x}^{\mathrm{E}}$ using Eq. (12); // Token embedding.
  else
    Map $\boldsymbol{x} \rightarrow \boldsymbol{x}^{\mathrm{E}}$ using Eq. (13); // Token conversion.
  end if
  Map $\boldsymbol{x}^{\mathrm{E}} \rightarrow \boldsymbol{x}$ using Eq. (14); // Transformer block with W-MSA.
  Map $\boldsymbol{x} \rightarrow \boldsymbol{x}$ using Eq. (14); // Transformer block with SW-MSA.
end for
Map $\boldsymbol{x} \rightarrow \boldsymbol{x}^{\mathrm{O}}$ using Eq. (18); // Reconstruct samples using latent representation.
Map $\boldsymbol{x}^{\mathrm{O}} \rightarrow \boldsymbol{X}^{\mathrm{R}}$ using Eq. (19). // Replacement.

$$\boldsymbol{Y}_{m}^{\hat{n}} = \left\{ y_{im}^{\hat{n}} \right\}_{i=1}^{T} \in \mathbb{R}^{T \times 1} \tag{5}$$

And the time series of all N turbines' m_{th} feature $\boldsymbol{Y}_{m}$ is defined as:

$$\boldsymbol{Y}_{m} = \left\{ \boldsymbol{Y}_{m}^{n} \right\}_{n=1}^{N} = \left\{ \boldsymbol{Y}_{im} \right\}_{i=1}^{T} \in \mathbb{R}^{T \times N} \tag{6}$$

To calculate the correlation coefficient between the $\hat{n}_{th}$ turbine and other turbines, a complete time series should be utilized. Consequently, time steps containing missing values should be omitted:

$$\boldsymbol{Y}_{m}^{n,\mathrm{C}} = \left\{ y_{im}^{n} | 0 < i \leq T, \boldsymbol{Y}_{im} \text{ is complete} \right\} \tag{7}$$

The Pearson correlation between the $\hat{n}_{th}$ turbine and the n_{th} turbine is calculated as follows:

$$r(\hat{n}, n) = \frac{\sum_{i} \left(y_{im}^{\hat{n},\mathrm{C}} - \bar{y}_{m}^{\hat{n},\mathrm{C}} \right) \left(y_{im}^{n,\mathrm{C}} - y_{m}^{n,\mathrm{C}} \right)}{\sqrt{\sum_{i} \left(y_{im}^{\hat{n},\mathrm{C}} - \bar{y}_{m}^{\hat{n},\mathrm{C}} \right)^{2}} \sqrt{\sum_{i} \left(y_{im}^{n,\mathrm{C}} - \bar{y}_{m}^{n,\mathrm{C}} \right)^{2}}} \tag{8}$$

where the overline $\overline{(\cdot)}$ signifies the mean value computed across all time steps.

To identify the most correlated turbines relative to the $\hat{n}_{th}$ turbine, the $\hat{K}$ largest correlation coefficients exceeding a predetermined threshold are retained, represented as $\{r(\hat{n}, n_1),\ r(\hat{n}, n_2),\ \ldots,\ r(\hat{n}, n_{\hat{K}})\}$, with $\{n_1,\ n_2,\ \ldots,\ n_{\hat{K}}\}$ representing the corresponding turbines. Subsequently, K turbines are randomly selected from the $\hat{K}$ turbines for combining feature representation. The random selection aims to increase the diversity of turbine combinations, providing flexibility and serving as a data augmentation technique if applied repeatedly. The combined features are created by concatenating all features at each time step of the selected turbines, including the $\hat{n}_{th}$ turbine, along the feature dimension. This can be expressed as:

$$\boldsymbol{X}^{\hat{n}} = \mathrm{concat}(\boldsymbol{Y}^{n_1}, \boldsymbol{Y}^{n_1}, \cdots,\ \boldsymbol{Y}^{n_K}, \boldsymbol{Y}^{\hat{n}}) \in \mathbb{R}^{T \times M(K+1)} \tag{9}$$

where M represents the number of features associated with each individual turbine. The process of feature combination is illustrated in Fig. 3.

It should be noted that although all features of an individual turbine within a group will be concatenated together, the choice of features for calculating correlation coefficients can impact the efficacy of feature combination. Wind speed measurement is advocated as the most indicative feature of a wind turbine's operational status with high volatility and stochastic nature. Therefore, wind speed is crucial for inferring other turbine features but presents a significant challenge when imputation is required due to missing data. Moreover, wind speed

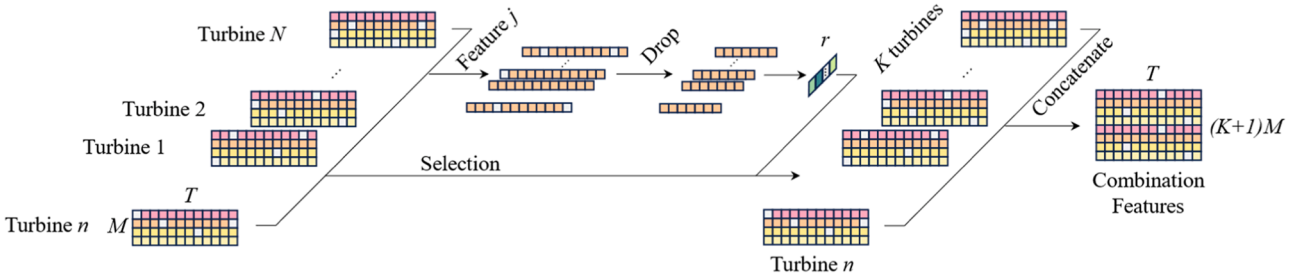

Fig. 3. Feature Combination.

is the most effective feature in capturing geographical correlations, making it the preferred choice for calculating correlation coefficients across turbines [45]. Features such as power output and temperature, while useful, are less effective than wind speed, as they can be derived from wind speed, exhibit lower variability, or may be subject to human influences.

3.3. Masking

Masking, a technique that explicitly compels the model to reconstruct masked data, renders the training process akin to imputation tasks. For each forward propagation of a time series sample $\boldsymbol{X}$, a masking matrix $\boldsymbol{M}$, possessing the same shape as $\boldsymbol{X}$, is generated based on a particular masking strategy. The masking matrix $\boldsymbol{M} = (m_{ij})$ represents the values to be masked, with the definition as follows:

$$m_{ij} = \begin{cases} 0 & \text{if } x_{ij} \text{ is observed} \\ 1 & \text{if } x_{ij} \text{ is to be masked} \end{cases} \tag{10}$$

The masking matrix closely resembles the missing matrix, hence, for the sake of simplicity and convenience, the same letter 'M' is employed to represent them.

Three masking strategies are introduced to align the masking process with data missing, encompassing random masking, band masking, and feature masking. These masking strategies generate mask matrixes that satisfy the definition of missing matrices for the respective missing types. To investigate a broad range of missing conditions, randomness is incorporated into the process by independently sampling masking positions for each masking procedure. Random masking is a straightforward approach that randomly selects a percentage of values following a uniform distribution. Band masking and feature masking involve two sampling stages. In the first stage, a turbine or a feature are randomly chosen. In the second stage, an initial masked time step is randomly chosen, along with subsequent time steps. These two sampling stages are independent of each other and follow uniform distributions. When applying band masking, all features of the sampled turbine in the selected time steps are masked, and when utilizing feature masking, the sampled feature of all turbines during the selected time period is masked. The masking ratio is employed to quantify the masking scale, with a definition equivalent to the missing ratio.

Once the masking matrixes generated, the masked values will be replaced with zeroes as follows:

$$\boldsymbol{X}^{\mathrm{M}} = \boldsymbol{X} \odot (\mathbf{1} - \boldsymbol{M}) \tag{11}$$

where $\mathbf{1}$ represents a matrix of ones with the same shape as $\boldsymbol{M}$.

3.4. Shifted window based multiscale attention

In various instances of missing data, the dominant imputation clues may differ in terms of correlation scales. To explicitly incorporate feature correlation, turbine correlation, and temporal correlation into a unified model, the MAMAE model employs a multiscale attention architecture derived from the transformer. Based on self-attention, transformer selectively blends information from different tokens in a sequence by calculating correlations between them. Traditional transformer models for time series handle sequences of time steps and treat each step as a token, limiting the self-attention mechanism to extracting time correlations only, but by altering the token's physical representation, the attention mechanism can extract different scales of correlations. Therefore, this work presents a multiscale attention method which is realized by reconstructing sequences and converting the physical representation of tokens within them.

The multiscale attention employs four distinct token types for diverse scale representation: feature token, turbine token, short-term token, and long-term token. A feature token represents a high-dimensional mapping of a single feature at a specific time step. In contrast, a turbine token encapsulates the high-dimensional mapping of all features associated with a specific turbine at a particular time step. The short-term token covers the mapping of all turbines' features within a single time step, while the long-term token spans across multiple time steps. The four types of tokens are processed individually in four respective transformer stages. In each transformer stage, the original sequence $\boldsymbol{X} \in \mathbb{R}^{T \times D}$, representing time steps, is first reconstructed by token embedding or token conversion module to generate a sequence of corresponding tokens. This token conversion process is illustrated in Fig. 4. Afterward, the self-attention mechanism is applied to the reconstructed sequence in subsequent transformer blocks to extracts correlations between tokens.

Although multiscale attention provides comprehensive correlation extraction by attending to diverse tokens, it concurrently involves computation overhead. In the feature and turbine scales, an original time step is divided into multiple tokens, resulting in a significantly increased sequence length. Since the traditional attention mechanism possesses quadratic computational complexity with respect to sequence length, this leads to considerable computation cost. For transformer-based models, the computational complexity is a critical consideration due to the high graphics processing unit (GPU) memory requirements inherent in training. A quadratic increase in computational complexity dramatically intensifies GPU memory consumption, potentially rendering the training process excessively costly. Consequently, window-based multihead self-attention (W-MSA) and shifted window-based multihead self-attention (SW-MSA) are introduced to substitute multihead self-attention (MSA) in conventional transformer blocks, forming window-based multiscale attention (W-MA) in this study. By applying attention mechanisms only in non-overlapping windows, W-MSA and SW-MSA exhibit linear rather than quadratic computational complexity (details in Section 3.4.3). The incorporation of W-MSA and SW-MSA effectively reduces the computation complexity associated with the first two attention scales.

3.4.1. Token embedding and conversion

The token embedding and conversion modules serve distinct purposes within the token conversion process. The token embedding module is specifically designed for embedding purposes, as employed in the first transformer stage. It comprises two reshaping functions, a concatenating function, and a fully connected layer. The two reshaping functions serve to generate sequences of feature tokens. Subsequently, a

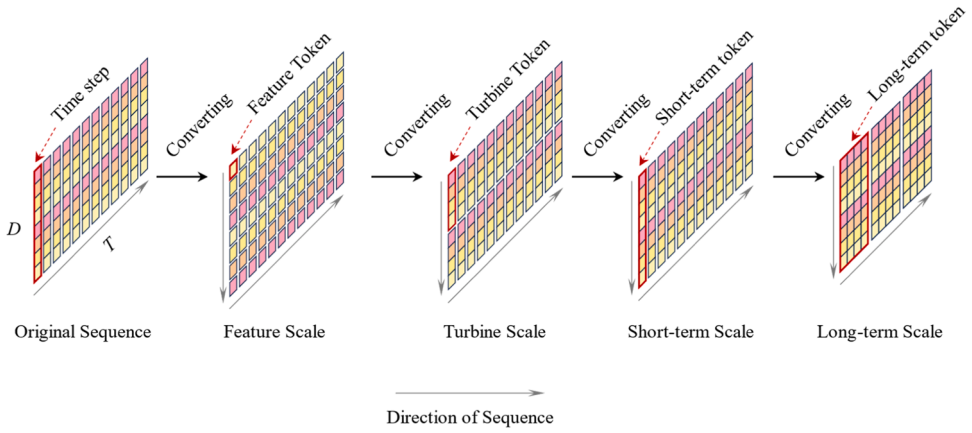

Fig. 4. Token conversion.

concatenating function is employed to indicate the position of masked values in the sequence, enabling the model to accurately identify them. Due to the presence of zeroes in the original data distribution, the model may struggle to discern between existing values and masked ones if the masked data are directly fed into the model. By concatenating $\boldsymbol{X}$ and $\boldsymbol{M}$ along the feature dimension, each original feature is projected into a two-dimensional space where one dimension describes the values and the other indicates whether the values are masked. In this space, existing and masked values are separated, and the separation facilitates easier differentiation between masked and existing values for the model. Finally, a fully connected layer is utilized to accomplish high-dimensional embedding of the input for subsequent transformer blocks. The token embedding can be represented as follows:

$$\begin{aligned}\boldsymbol{x} &= \text{reshape}\left(\boldsymbol{X},\ L^{\text{F}}, 1\right)\\ \boldsymbol{m} &= \text{reshape}\left(\boldsymbol{M}, L^{\text{F}}, 1\right)\\ \widehat{\boldsymbol{x}} &= \text{concat}(\boldsymbol{x}, \boldsymbol{m})\\ \boldsymbol{x}^{\text{E}} &= \widehat{\boldsymbol{x}}\boldsymbol{W}^{\text{E}} + \boldsymbol{b}^{\text{E}}\end{aligned} \tag{12}$$

where L^{F} denotes the length of the feature scale sequence, $\text{reshape}\left(\boldsymbol{X}, L^{F}, 1\right)$ indicates that the matrix $\boldsymbol{X}$ is to be transformed into a new matrix with L rows and 1 column. and both $\boldsymbol{W}^{\text{E}}$ and $\boldsymbol{b}^{\text{E}}$ are learnable parameters of the fully connected layer.

The token conversion module comprises a reshape function, a layer normalization layer [46], and a linear projection. The computation process of the token conversion module is as follows:

$$\begin{aligned}\widehat{\boldsymbol{x}} &= \text{reshape}\left(\boldsymbol{x},\ L^{\text{R}}, D^{\text{R}}\right)\\ \boldsymbol{x}^{\text{E}} &= \text{LN}(\widehat{\boldsymbol{x}})\boldsymbol{W}^{\text{R}}\end{aligned} \tag{13}$$

where $\boldsymbol{x}$ denotes the input of the token conversion module, while L^{R} and D^{R} signify the length of the reshaped sequence and the feature dimension of tokens, respectively. LN denotes the layer normalization, and $\boldsymbol{W}^{\text{R}} \in \mathbb{R}^{D^{\text{R}} \times D^{\text{E}}}$ is a learnable parameter. Distinct from the fully connected layer in token embedding, the linear projection in the token conversion serves to compress the input with the D^{E} equal to $D^{\text{R}}/2$.

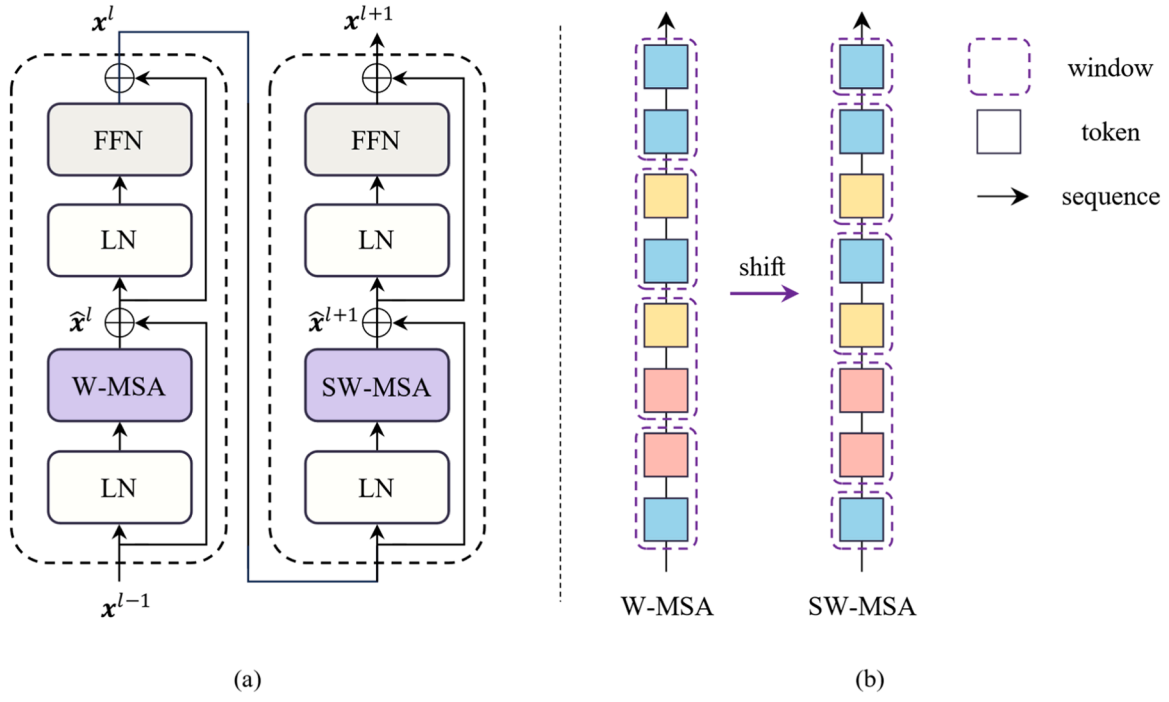

Fig. 5. Transformer with W-MSA and SW-MSA. (a) Structure of two successive transformer blocks. (b) window shifting of successive W-MSA and SW-MSA.

3.4.2. Transformer block

In this work, the transformer block is designed based on the W-MSA and the SW-MSA to decrease computational complexity. The structure is predominantly consistent with that of Ref. [47], except for the modifications made to the (S)W-MSA for 1-dimensional sequences. As shown in Fig. 5(a), transformer blocks occur in pairs, with one containing W-MSA and the other SW-MSA (justification provided in Section 3.4.3). A single transformer block comprises two layer normalizations, a (S)W-MSA layer, and a position-wise feed-forward network (FFN). The FFN is a two-layer multilayer perceptron (MLP) with a gaussian error linear units (GELU) activation [48] in between. This network operates independently on each token while sharing the same parameters across tokens. Before each (S)W-MSA and FFN layer, a layer normalization is applied, followed by a residual connection as is typical in most transformer models. In summary, a transformer block can be expressed as follows:

$$\begin{aligned}\hat{\boldsymbol{x}}^{l} &= (\mathrm{S})\mathrm{W}-\mathrm{MSA}\left(\mathrm{LN}\left(\boldsymbol{x}^{l-1}\right)\right)+\boldsymbol{x}^{l-1}\\ \boldsymbol{x}^{l} &= \mathrm{FFN}\left(\mathrm{LN}\left(\hat{\boldsymbol{x}}^{l}\right)\right)+\hat{\boldsymbol{x}}^{l}\end{aligned} \tag{14}$$

where the $\hat{\boldsymbol{x}}^{l}$ and $\boldsymbol{x}^{l}$ denote the output of (S)W-MSA and FFN in the l_{th} block, respectively.

3.4.3. (Shifted) window based multihead self-attention

The W-MSA employs self-attention within local windows rather than the entire sequence, resulting in a substantial reduction in computational complexity. However, this approach lacks communication across windows. To overcome this limitation and maintain the efficiency of window-based calculations, a subsequent SW-MSA is introduced. As illustrated in Fig. 5(b), the window in the SW-MSA is shifted by half of the window size compared to that in the W-MSA. By implementing successive W-MSA and SW-MSA, the attention mechanism can be applied on the connections between respective local windows, thereby facilitating the efficient extraction of cross-window information. Moreover, as (S)W-MSA is implemented across all attention scales, the high-level receptive field encompassing multiple low-level windows alleviates the constraint imposed by window partitioning.

Self-attention is insensitive to position, rendering it unable to leverage the sequence's order. Consequently, a learnable relative position bias is introduced for tokens within a window when computing attention weights. Assuming a window size denoted by w, the distance between two tokens in the window along the sequence ranges from $[-w+1, w-1]$, which implies $(2w-1)$ possible relative positions. To this end, a learnable relative position vector $\widehat{\boldsymbol{B}}=(\widehat{b}_i)\in\mathbb{R}^{2w-1}$ is introduced for the relative position bias represented by $\boldsymbol{B}=(b_{ij})\in\mathbb{R}^{w\times w}$. The relative position bias is derived from the relative position vector through a relative position index. The generation of the relative position bias can be expressed as follows:

$$b_{ij}=\widehat{\boldsymbol{B}}_{(j-i+w)} \tag{15}$$

with the presence of relative position bias, the calculation of multihead self-attention within a single window proceeds as follows:

$$\begin{aligned}\boldsymbol{Q}_i &= \boldsymbol{x}\boldsymbol{W}_i^{\mathrm{Q}}\\ \boldsymbol{K}_i &= \boldsymbol{x}\boldsymbol{W}_i^{\mathrm{K}}\\ \boldsymbol{V}_i &= \boldsymbol{x}\boldsymbol{W}_i^{\mathrm{V}}\\ \boldsymbol{head}_i &= \mathrm{softmax}\left(\frac{\boldsymbol{Q}_i\boldsymbol{K}_i^{T}}{\sqrt{d_h}}+\boldsymbol{B}\right)\boldsymbol{V}\\ \mathrm{MSA}(\boldsymbol{x}) &= \mathrm{concat}(\boldsymbol{head}_1,\cdots,\boldsymbol{head}_h)\boldsymbol{W}^{\mathrm{O}}\end{aligned} \tag{16}$$

where $\boldsymbol{x}\in\mathbb{R}^{w\times D^{\mathrm{E}}}$ represents the input sequence within a window, $\boldsymbol{W}\in\mathbb{R}^{D^{\mathrm{E}}\times d_{\mathrm{h}}}$ denotes the learnable parameters, d_{h} signifies the embedding dimension for $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$, and h refers to the number of heads. Note that d_{h} is equivalent to D^{E} divided by h.

The computational complexity can be campared between the global MSA and the (S)W-MSA. Assuming that MSA is applied to a sequence of length l with d features, the complexity of the global MSA and (S)W-MSA respectively amounts to:

$$\begin{aligned}O(\mathrm{MSA}) &= 4ld^2+2l^2d\\ O(\mathrm{W}-\mathrm{MSA}) &= 4ld^2+2lwd\end{aligned} \tag{17}$$

where the former exhibits a quadratic relationship with sequence length l, whereas the latter reduced to linear. This inherent strength enables efficient computation at both the feature and turbine scales.

3.5. Prediction

Lastly, a prediction module is incorporated to generate imputation outcomes utilizing the latent representation extracted by the multiscale attention mechanism. The prediction module comprises layer normalization, a linear layer with bias for low-dimensional projection, and a reshape function to restore the original sequence shape. The prediction module can be represented as follows:

$$\begin{aligned}\hat{\boldsymbol{x}} &= \mathrm{LN}(\boldsymbol{x})\boldsymbol{W}^{\mathrm{P}}+\boldsymbol{b}^{\mathrm{P}}\\ \boldsymbol{x}^{\mathrm{O}} &= \mathrm{reshape}(\hat{\boldsymbol{x}},T,D)\end{aligned} \tag{18}$$

where T and D denote the sequence length and feature dimension of the original input, respectively. During the imputing process, a replacement strategy is utilized as illustrated below:

$$\boldsymbol{X}^{\mathrm{R}}=\boldsymbol{M}\odot\boldsymbol{x}^{\mathrm{O}}+(\boldsymbol{1}-\boldsymbol{M})\odot\boldsymbol{X}^{\mathrm{I}} \tag{19}$$

The replacement strategy aims to retain existing values unchanged, a feat that cannot be guaranteed by neural networks. Nonetheless, during the training process, no values in the output will be replaced for the sake of easier convergence.

4. Experiments

4.1. Dataset and preprocessing

A real-world SCADA dataset, collected from an offshore wind farm in China, was utilized in the experiments aimed at validating the performance of the proposed method. The dataset encompasses 51 turbines and 2 measurements (wind speed and wind power), recorded at 10-minute intervals over a span of 1 year. To eliminate the influence of outliers, the data cleaning method from Ref. [49], incorporating a cubic polynomial-based smoothing technique, was applied to the raw data. An example of data cleaning is provided in Fig. 6(a). The dataset was subsequently divided into training, validation, and test sets, each containing 30, 10, and 11 wind turbines, respectively. This partitioning prevents cross-dataset information leakage.

Correlation coefficients of wind speeds across all turbines were calculated as depicted in Fig. 6(b), and feature combination was applied to each of the three datasets based on these results. With a threshold of 0.92 and K equal to 3, 6 random choices were made to select the highly-correlated turbines. If $\widehat{K}$ was less than 3 for a target turbine, the three turbines with the highest correlation coefficients were selected without random choice. For the combined features, a time step was considered complete only if it was complete for all turbines in the group. The number of turbine groups in each dataset is 174, 54, and 66, respectively.

In order to keep consistent with the input and output requirements of the model, time series datasets were generated. Additionally, for comparison purposes, datasets without feature combination were also developed. A sliding window of 96 time steps with a half-length sliding interval was applied to each dataset. To conduct a comprehensive evaluation, it is essential to utilize existing and accurate data, ensuring the credibility of the experimental outcomes. Therefore, for incomplete

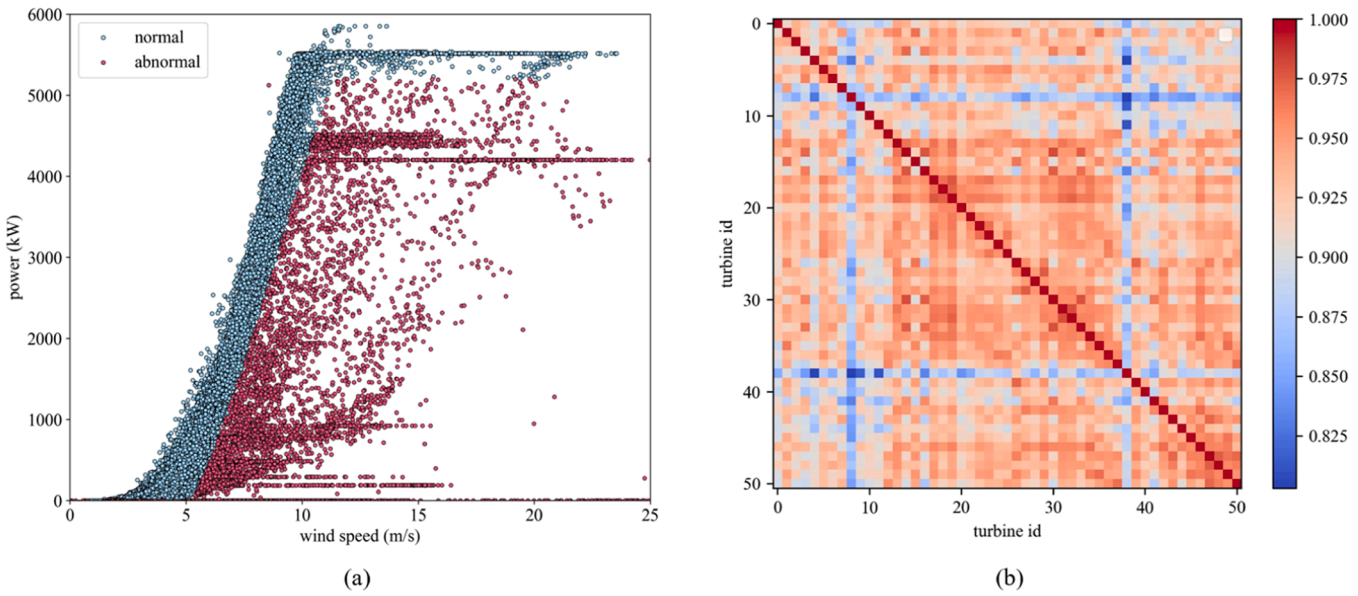

Fig. 6. Preprocessing. (a) An example of data cleaning. (b) Wind speed correlation coefficient.

time series, if the percentage of missing time steps exceeded 5 %, the series would be discarded. Otherwise, the missing values were replaced using linear interpolation with existing data. Artificial masking was employed in the test set to simulate missing data for evaluation, aligning with the three identified missing types. The number of samples in each dataset with feature combination is 7639, 3796, and 2249 respectively. For datasets without feature combination, the training and validation sets contain 7945 and 2706 samples respectively. To guarantee an equitable comparison, the test set was derived from the dataset with feature combination, exclusively containing features specific to the target turbine. Standard normalization was implemented on all datasets using identical parameters.

4.2. Experiments settings

Two experiments were conducted in this paper. The first experiment evaluated the accuracy of the MAMAE in various missing data situations and compared the results with other prevalent methods, using the datasets with feature combination. The missing data situations encompassed three distinct missing types, and nine distinct missing ratios were considered for each type. Seven prevalent imputation methods were employed as benchmarks. Additionally, the MAMAE's generalization performance was assessed by evaluating all three masking strategies and nine masking ratios for each missing situation. The second experiment served as an ablation study to evaluate the proposed modifications: the multiscale attention, (S)W-MSA, feature combination, and masked autoencoding. The effectiveness of these modifications was assessed utilizing accuracy and efficiency metrics to determine their contributions to the MAMAE framework. The ablation experiment was conducted using both kinds of datasets, incorporating and excluding feature combination.

Two metrics were used for accuracy and efficiency assessment, respectively. For assessing the imputation accuracy, the mean squared error (MSE) between the missing and the imputed values was employed as the evaluation metric. The computation of the MSE is represented below:

$$\text{masked-MSE} = \frac{\sum_{n=1}^{N}\sum_{i=1}^{T}\sum_{j=1}^{D}\left(x_{ij}^{\mathrm{I},n} - x_{ij}^{\mathrm{R},n}\right)^2 m_{ij}^{\mathrm{I},n}}{\sum_{n=1}^{N}\sum_{i=1}^{T}\sum_{j=1}^{D} m_{ij}^{\mathrm{I},n}} \tag{20}$$

where N denotes the number of samples, $\boldsymbol{X}^{\mathrm{I},n}$, $\boldsymbol{X}^{\mathrm{R},n}$, and $\boldsymbol{M}^{\mathrm{I},n}$ denote the n_{th} sample of original incomplete data, reconstructed data and masking matrixes respectively. Meanwhile, the floating point operations (FLOPs) was adopted in the ablation experiments as a metric of the computational complexity to assess the efficacy of the (S)W-MSA in enhancing efficiency.

Clarification of the model configurations is provided as follows. The proposed MAMAE involves a limited number of independent structural hyperparameters, including the initial embedding dimension d_{embed}, window size w, the number of heads h, the hidden size of the FFN d_{hidden}, and the drop rate p_{drop}. To simplify the configuration, the hidden size of each FFN was set to a specific multiple n_{hidden} of its respective input size, while maintaining an output size equal to the input size. The drop rate is a hyperparameter in the drop-path [50] which is applied to the shortcut of the residual connection in each transformer block. The drop-path randomly sets the values in a certain proportion of batches to zeroes, contributing to the prevention of overfitting. The model was trained using the Adam optimizer [51] for 200 epochs, employing a cosine decay [52] learning rate schedule. The learning rate varies with the number of epochs as follows:

$$lr(t) = \begin{cases} \dfrac{t}{T_0}\left(lr_{\text{init}} - lr_{\text{inf}}\right) + lr_{\text{inf}} & \text{if } t \le T_0 \\ \max\left\{lr_{\text{inf}}, \dfrac{1}{2}\left[1 + \cos\dfrac{\pi(t - T_0)}{T - T_0}\right] lr_{\text{init}}\right\} & \text{if } t > T_0 \end{cases} \tag{21}$$

where t represents the current epoch, T_0 denotes the epochs for warm-up, T represents the total number of epochs, and lr stands for the learning rate. The main hyperparameters were determined through experience and random search. The final hyperparameters are enumerated in Table 1. For the training loss function, the mean squared error (MSE) was computed over all values, rather than masked ones:

Table 1
Hyperparameter setting.

Config	value	config	value
d_{embed}	64	lr_{init}	2.0×10^{-4}
w	12	lr_{inf}	2.0×10^{-5}
h	8	T_0	20
n_{hidden}	4	T	200
p_{drop}	0.1	batch size	128

$$\mathrm{MSE} = \frac{1}{NTD}\sum_{n=1}^{N}\sum_{i=1}^{T}\sum_{j=1}^{D}\left(x_{ij}^{\mathrm{I},n} - x_{ij}^{\mathrm{R},n}\right)^2 \tag{22}$$

It was found easier for convergence of training when calculating MSE over all values.

Seven benchmarks are implemented in the experiments to verify the effectiveness of the proposed MAMAE:

(i) The linear interpolation: an interpolation-based method. It calculates the linear combination of the two time steps before and after the time steps with missing values.
(ii) The mean value interpolation: a similarity-based method. It replaces the missing values with the mean value of the corresponding feature.
(iii) The k-nearest neighbors (KNN): a similarity-based method. It calculates the distance between the incomplete and complete samples, chooses k nearest samples and average the selected samples to impute incomplete samples. Considering that the KNN assesses similarity solely based on the features of a single time step, a dataset composed of isolated time steps was thereby generated as the training set for the KNN. Due to the unaffordable computational cost imposed by a vast number of training samples, the dataset randomly sampled 10,000 time steps from the training set of time series. The number of nearest neighbors employed for imputation was set to 10.
(iv) The autoencoder (AE): an autoencoding model. A 4-layer MLP with two layers for compression and the remaining layers for reconstruction are implemented. Each compression layer features a hidden size approximately 1.2 times larger than the subsequent layer, creating a bottleneck with 533 hidden units – approximately 1.5 times shrinking. The hidden sizes of the successive reconstruction layers follow a consistent growth multiplier, ensuring output size equality to the input size. All structural hyperparameters and training configurations were determined via random search.
(v) The bidirectional recurrent imputation for time series (BRITS) [21]: an autoregression deep learning method. It implements bidirectional autoregression by using two LSTM-based unidirectional recurrent dynamics.
(vi) The generative adversarial imputation nets (GAIN) [53]: an adversarial training deep learning method. It consists of a discriminator and a generator, using the data generated by generators to impute the incomplete sample.
(vii) The self-attention-based imputation for time series (SAITS) [29]: a deep learning method integrating masked autoencoding. It is based on diagonally-masked self-attention and transformer architecture, imputing the missing values using a learnable weighted averaging of the outputs from two successive diagonally-masked self-attention (DSMA) blocks.

All deep learning methods were realized by PyTorch [54] while the KNN and linear interpolation were implemented via scikit-learning [55] in Python. All experiments results were conducted on NVIDIA GeForce RTX 3090 24GB and Intel(R) Xeon(R) CPU E5–2680 v4 @ 2.40 GHz.

4.3. Imputation performance

4.3.1. Random missing

The masked-MSE of various methods across different random missing ratios is summarized in Table 2, with the optimal metrics highlighted in bold. The error distributions of 20 % (low missing ratio), 50 % (medium missing ratio) and 80 % (high missing ratio) random missing are presented in Fig. 7. The MAMAE was trained utilizing the same masking types and ratios as the missing conditions. It is noteworthy that at a missing ratio of 90 %, the linear interpolation lacks sufficient data to perform missing value imputation, resulting in no corresponding outcome.

The proposed MAMAE model exhibits optimal accuracy in scenarios characterized by high missing ratios. Conversely, in situations with low missing ratios, the SAITS model outperforms the MAMAE. This can be attributed to SAITS's implementation of diagonally-masked self-attention and its weighted-combination design. The former approach discounts intra-step correlations, compelling the model to infer inter-step relationships, a strategy that proves more efficient than conventional self-attention for imputing missing data. The latter design involves assigning weights to the outputs of two DSMA blocks, thereby enhancing SAITS's adaptability to diverse incomplete samples. Nevertheless, the SAITS is less precise than the MAMAE when confronting large-scale missing. These findings suggest that multiscale correlations are of greater importance in extensive data missing scenarios than singular temporal correlations. Although the training tasks of the SAITS and the MAMAE are akin, the SAITS's attention mechanism remains confined to temporal correlations, unlike the explicit multiscale application in MAMAE.

Among the other comparative methods, the BRITS emerges as the superior option, particularly as the only RNN-based approach. The BRITS, akin to the SAITS, can effectively extract temporal correlations but lacks a dedicated mechanism for discerning other types of dependencies. However, the BRITS underperforms the SAITS due to the lesser potency of its autoregression compared to masked autoencoding. Autoregression's reliance on time series prediction lacks a global receptive field and encounters error compounding. The two MLP-based methods, GAIN and AE, both lack an efficient mechanism for extracting multiscale correlations. Moreover, their training tasks are not as potent as those employed in masked autoencoding. The GAIN, as an adversarial training method, is found to struggle with convergence challenges. The AE, by nature, incurs information loss due to its bottleneck architecture and experiences reconstruction errors even with complete samples. Additionally, it fails to generalize to input data distributions that include large-scale missing values, leading to a rapid decline in accuracy as the missing ratio escalates. Traditional imputation methods, such as linear interpolation and KNN, as mentioned in the Introduction section, are ineffective in capturing correlations.

In general, the MAMAE model tends to outperform most benchmarks when confronted with randomly missing data due to its capability for

Table 2
Masked-MSE in random missing condition.

Model	Missing Ratio								
	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %
MAMAE	0.013	0.015	0.017	0.019	0.022	**0.025**	**0.029**	**0.038**	**0.055**
SAITS	**0.010**	**0.012**	**0.014**	**0.017**	**0.020**	0.026	0.035	0.063	0.172
BRITS	0.027	0.030	0.034	0.038	0.042	0.047	0.055	0.069	0.308
GAIN	0.060	0.077	0.094	0.109	0.123	0.145	0.258	0.352	1.734
Interpolation	0.030	0.034	0.038	0.045	0.054	0.069	0.099	0.172	\
KNN	0.051	0.057	0.064	0.074	0.091	0.125	0.191	0.326	0.588
AE	0.077	0.114	0.169	0.240	0.329	0.434	0.559	0.702	0.865
Mean	0.275	0.274	0.275	0.275	0.277	0.279	0.281	0.287	0.303

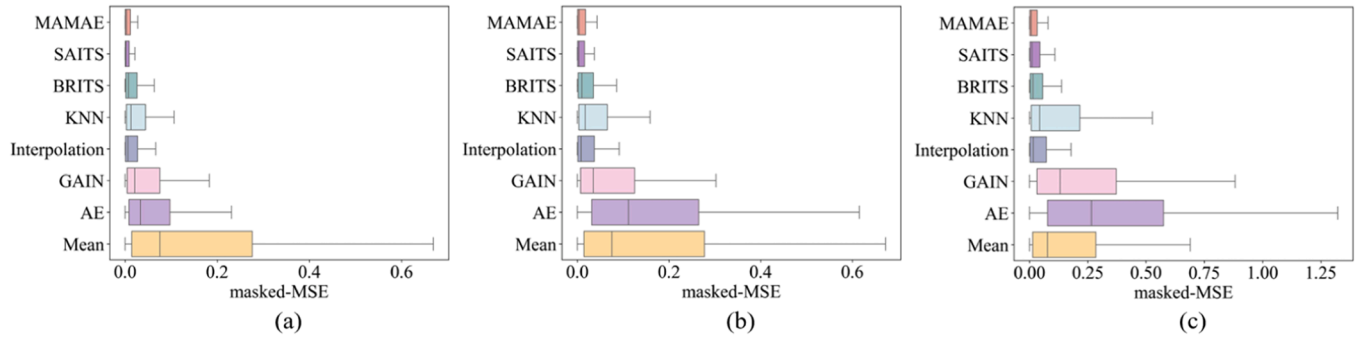

Fig. 7. Error distributions. (a) 20 % random missing; (b) 50 % random missing; (c) 80 % random missing.

comprehensively extracting correlations and its powerful training task. Owing to its multiscale attention architecture, the MAMAE surpasses the SAITS in scenarios with extensive missing data. However, it may underperform in cases of sparse missing values, lacking the benefits of SAITS's diagonally-masked self-attention and weighted-combination design. However, although random missing serves as a benchmark in most imputation studies, it is not common in SCADA data. Therefore, the performance of the MAMAE should be further evaluated in band and feature missing scenarios.

4.3.2. *Band missing*

The results for band missing are presented in Table 3, in which the MAMAE achieves the highest accuracy across most scenarios. Moreover, the error distributions are presented in Fig. 8. Notably, the performance of the SAITS deteriorates, outperforming the MAMAE only in the 10 % missing scenario. Considering the inherent randomness in wind speed variations, band missing, characterized as a form of continuous missing, necessitates the exploitation of turbine correlations over temporal correlations for effective imputation. Consequently, the MAMAE's ability to extract multiscale correlations becomes particularly advantageous. Furthermore, the masking strategy employed by the SAITS is tailored for common random missing and does not account for the distinctive characteristics of band missing or feature missing. This oversight leads to a marked decline in accuracy as the missing ratio increases. The influence of masking strategy will be further discussed in Section 4.3.4. The KNN performs well in high missing ratio scenarios, outperforming both SAITS and BRITS, with its accuracy remaining nearly constant despite variations in the missing ratio. This is attributed to KNN's disregard for temporal correlations, instead relying solely on other features within the same time step for imputation. As the missing ratio increases and the extent of continuous missing time steps grows, KNN's accuracy remains unaffected. However, as KNN's correlation extraction capability is inferior to that of MAMAE, it does not surpass MAMAE consistently. Another phenomenon distinct from the random missing scenario is the significant degradation in performance of the linear interpolation as the missing ratio increases, underscoring the limitation of relying solely on single correlations and highlighting the necessity of the multiscale attention architecture in MAMAE.

The remaining results and their underlying reasons align with those observed in the random missing scenario. The outcomes for band missing underscore the benefits of the enhancements introduced in the MAMAE model: advanced training tasks and comprehensive correlation extraction. These two benefits are particularly pronounced in the context of band missing, which are more prevalent in wind turbine SCADA datasets.

Fig. 9 illustrates several instances of imputation results under a 50 % missing data situation. For the sake of clarity, the methods presented here are limited to the MAMAE, the top three performing competitors, and the linear interpolation. Each subfigure presents a measurement in a time series sample comprising both existing and missing values. Since the missing values are simulated by artificially masking, the true values of the missing positions are known. Each subfigure presents the true values of the missing positions, along with the imputed values obtained from various methods, and the existing values. The MAMAE model most effectively captures the underlying trends of the true values.

4.3.3. *Feature missing*

The results of feature missing are demonstrated in Table 4 and the error distributions are presented in Fig. 10, which exhibits a similar performance order of the presented methods as the band missing. The discrepancies between the results of the two missing types deserve special attention. Firstly, the MAMAE consistently delivers the best performance across all missing ratios, providing compelling evidence of the model's two key strengths. Secondly, the masked-MSE in feature missing situation is lower than in the band missing situation. Although both missing types involve feature-led imputation tasks, the results suggest that utilizing correlated features within the same turbine offers better results than employing correlations from distinct turbines. Therefore, this highlights the preference for correlation within the same turbine over cross-turbine correlations in imputation tasks. Finally, the masked-MSE of the MAMAE barely increases as the missing ratio escalates from 10 % to 60 %, similar to the KNN. This characteristic indicates the diminished importance of time correlations in the feature missing. This phenomenon is not observed in band missing scenarios, as turbine correlations aid in acquiring the trend of attributes, but the exact values should be ascertained through the previous and subsequent time steps.

Table 3
Masked-MSE in band missing condition.

Model	Missing Ratio								
	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %
MAMAE	0.058	**0.050**	**0.056**	**0.056**	**0.057**	**0.058**	**0.061**	**0.062**	**0.064**
SAITS	**0.043**	0.063	0.081	0.099	0.118	0.136	0.170	0.201	0.271
BRITS	0.061	0.077	0.089	0.093	0.098	0.103	0.107	0.111	0.119
GAIN	0.135	0.164	0.186	0.200	0.237	0.260	0.345	0.412	0.496
Interpolation	0.102	0.334	0.936	0.584	1.865	2.603	4.310	8.871	23.756
KNN	0.097	0.096	0.099	0.096	0.095	0.096	0.096	0.096	0.097
AE	0.329	0.528	0.612	0.665	0.707	0.738	0.775	0.796	0.845
Mean	0.293	0.316	0.348	0.374	0.398	0.423	0.456	0.483	0.509

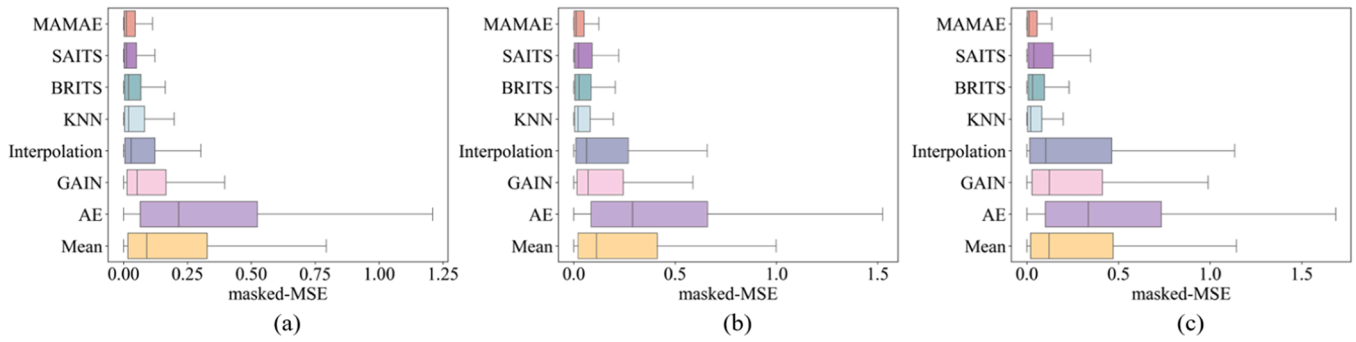

Fig. 8. Error distributions. (a) 20 % band missing; (b) 50 % band missing; (c) 80 % band missing.

In contrast, in the feature missing situation, the exact values of the missing data can be ascertained with little error using the attributes from the same turbine. Therefore, as the number of missing steps escalates, the performance of the MAMAE remains virtually unaltered. Several examples of 50 % feature missing are illustrated in Fig. 11.

4.3.4. Generalization of masking strategy

The adaptability of the masking strategy allows the MAMAE to accommodate various data distributions in distinct missing situations, which is crucial for its superior performance. However, the flexibility also causes training complexity. Although the missing conditions are known for a period of data, it is inevitable to train multiple models for diverse missing cases. If a model trained with a certain masking strategy can be utilized across a broad range of missing situations, the necessity for an extensive collection of trained models can be substantially diminished. In previous sections, the MAMAE employed the same masking type and ratio as the missing data; however, this section evaluates the generalization of the masking strategy to various missing cases. The experiments were initially conducted within the same masking type and then diversify among different masking types.

The masked-MSE for identical masking and missing types but varying masking and missing ratios are presented in Tables 5, 6, and 7, respectively for the random, band, and feature type masking. Additionally, the visualization is provided in Fig. 12.

The results indicates that the masking ratio exerts a considerable influence on the model's performance. The optimal training strategy does not involve maintaining a masking ratio identical to the missing ratio. Instead, the ideal masking ratio typically exceeds the missing ratio by a small margin, exhibiting an upward trend as the missing ratio increases. This phenomenon may be attributed to the fact that a larger masking ratio encourages the model to extract more information from the data. However, an excessively large discrepancy between the missing and masking ratios can alter the data distribution, resulting in deteriorated performance. Notably, the model exhibits a satisfactory generalization capability, as evidenced by its robust performance on a large scale of masking ratios. For both random and band missing, the performance metrics remain relatively constant within a range of approximately 10 % below to 10 % above the optimal masking ratio. For feature missing, the acceptable range is even broader. This suggests that in practical applications, the pursuit of an optimal training strategy is unnecessary.

To demonstrate cross-type generalization, three missing ratios, namely 20 %, 50 %, and 80 %, are selected. The model was trained with the same masking ratio but different masking types. The results are presented in Tables 8, 9, and 10, with visualizations provided in Fig. 13. The results show that the mismatching of masking type and missing type leads to significant performance degradation, indicating the lack of cross-type generalization. Among the masking types, random masking exhibits greater adaptability to various missing types, as it partially encompasses the patterns of band and feature missing. However, the resulting accuracy loss is still considerable.

The results indicate that while the same-type generalization of the masking strategy exhibits promising results, its cross-type generalization remains unsatisfactory. Notwithstanding, it is convenient to discern a particular missing type from the others, and the trio of missing types investigated in these experiments encompass the majority of potential missing situations, encompassing both discontinuous and continuous missing. For each missing type, as the model trained with a specific masking ratio has been proven capable of accommodating a broad range of missing ratios, it is unnecessary to train models for each possible missing ratio. Consequently, in real-world applications, it is sufficient to train a mere few models employing disparate masking strategies.

4.4. Ablation experiments

Ablation experiments were conducted to demonstrate the effectiveness of the proposed modifications. These experiments involved removing different modules from the model while keeping all other components unchanged. Specifically, four modules were sequentially removed, including the (S)W-MSA, the multiscale attention (MA), the feature combination (FC), and the masking. In the (S)W-MSA-removed model, the window size was set equal to the sequence length. In the MA-removed model, transformer blocks were directly applied to the high-dimensional embedding of the original sequence without any token conversion, and the initial embedding dimension was set as 128 to keep consistent with the embedding dimension in the short-term scale of the MAMAE. For the FC-removed model, the features of a single turbine were taken as input, using the single turbine dataset. The masking-removed model was also evaluated, but the results shows that the masked-MSE approximated the standard deviation under all conditions, indicating that the model has no imputation ability without masking. Therefore, the masking-removed model was not considered in the subsequent experiments.

The accuracy and computational complexity of the models were comprehensively compared using the masked-MSE and the FLOPs, respectively. Notably, to mitigate the influence of training strategy, the accuracy of the optimal masking strategies was examined. Given that the training strategy does not impact the FLOPs, models trained with 10 % random masking were selected for evaluation. The accuracy results are presented in Fig. 14, while Fig. 15 depicts the FLOPs. Regarding the multiscale attention, its explicit application across multiple scales to extract relevant correlations renders it superior to the MA-removed model, which solely focuses on the temporal scale. This is especially pronounced in the small-scale random missing and all-scale feature missing situations, where the baseline clearly outperforms the MA-removed model. However, it should be noted that with the increase in sequence length, the computational complexity, as measured by FLOPs, experiences a considerable increase if the (S)W-MSA is not utilized to replace the global attention. By incorporating the (S)W-MSA, although the computational complexity remains higher than that of the MA-removed model, it achieves a 17 % reduction in FLOPs compared to the (S)W-MSA-removed model, with a minimal impact on accuracy. This

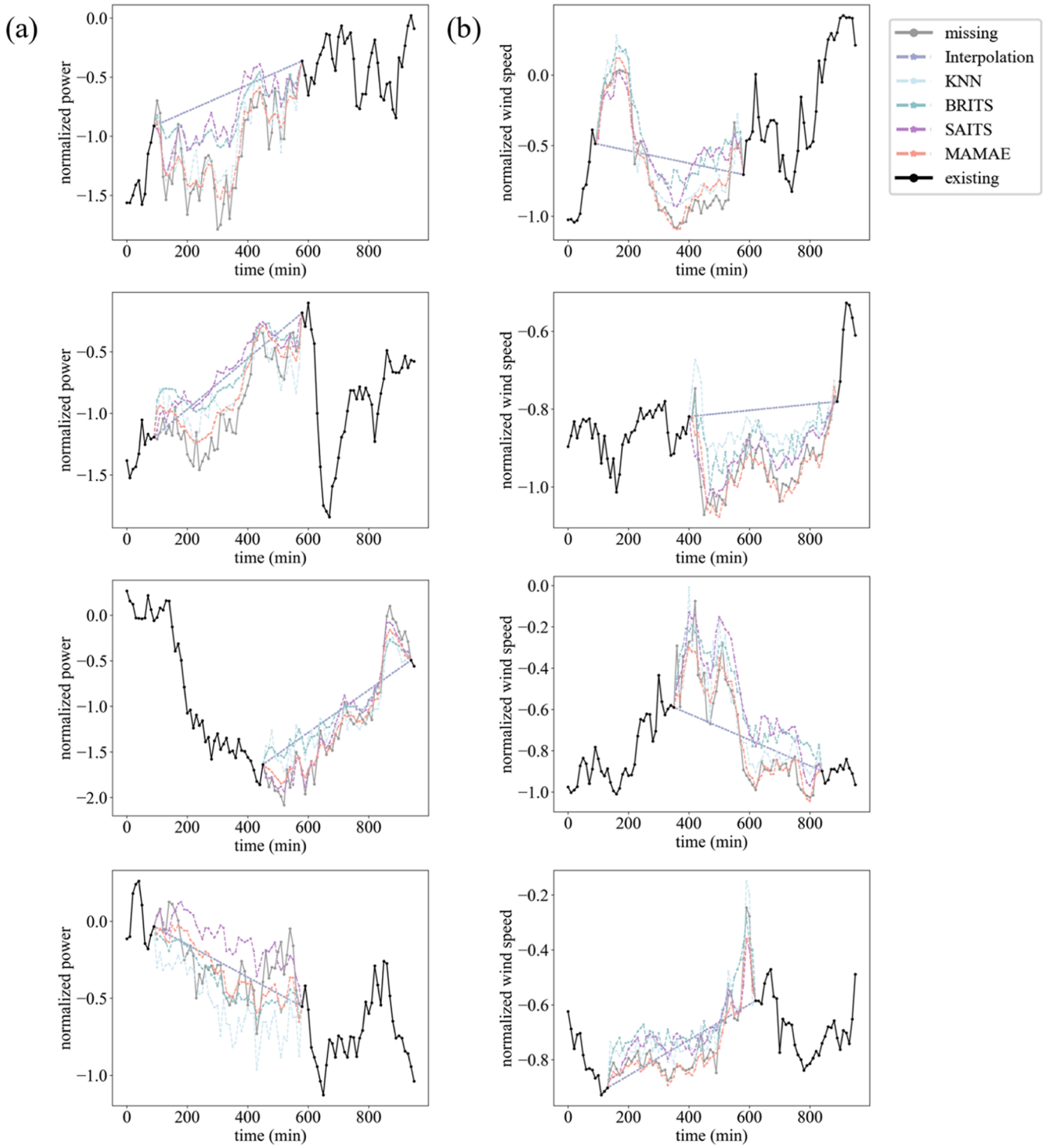

Fig. 9. Imputing results of 50 % band missing. (a) Normalized wind speed. (b) Normalized power.

Table 4
Masked-MSE in feature missing condition.

Model	Missing Ratio								
	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %
MAMAE	**0.023**	**0.020**	**0.021**	**0.021**	**0.022**	**0.023**	**0.026**	**0.026**	**0.028**
SAITS	0.028	0.046	0.040	0.064	0.071	0.094	0.129	0.167	0.227
BRITS	0.068	0.084	0.094	0.113	0.114	0.122	0.130	0.132	0.140
GAIN	0.151	0.239	0.311	0.427	0.563	0.797	1.049	1.137	1.387
Interpolation	0.115	0.298	0.811	1.042	1.930	2.960	6.329	9.756	20.833
KNN	0.077	0.080	0.074	0.077	0.074	0.075	0.078	0.078	0.080
AE	0.298	0.422	0.476	0.546	0.586	0.624	0.656	0.679	0.711
Mean	0.297	0.318	0.352	0.380	0.404	0.429	0.462	0.489	0.515

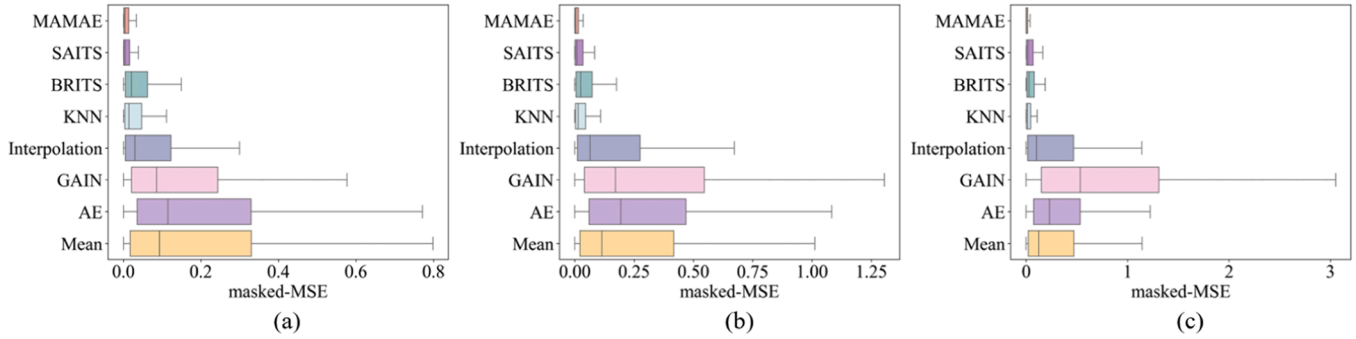

Fig. 10. Error distributions. (a) 20 % feature missing; (b) 50 % feature missing; (c) 80 % feature missing.

demonstrates a more cost-effective approach to implementing multiscale attention. As for the feature combination, despite potentially introducing redundancy when feature correlation is sufficient for imputation, facilitates the incorporation of turbine correlations, thereby substantially enhancing performance in the band missing and large-scale random missing cases. In summary, the multiscale attention and feature combination significantly improves accuracy, while the (S)W-MSA offers a reduction in computational cost with negligible performance decrement.

5. Discussion

In this section, imputation performance experiments and ablation studies were conducted to showcase the efficacy of the proposed MAMAE. The findings indicate that: (i) Owing to the dedicated training task for imputation and comprehensive capability for extracting correlations, the MAMAE exhibits superior accuracy compared to benchmarks across most missing data situations, particularly in the band missing and the feature missing which are more prevalent in wind turbine SCADA data than the random missing. (ii) The variance in performance under different missing data conditions highlights that the significance of various correlations varies with the missing patterns. This variability underscores the rationale behind the proposed multiscale attention mechanism. (iii) The training task exhibits strong same-type generalization capacity, suggesting that only a limited number of models need to be trained to cover a range of missing situations. (iv) The introduced modifications yield positive outcomes. The multiscale attention and feature concatenation contribute to enhanced accuracy, while the (S)W-MSA facilitates computational efficiency.

In this paper, the MAMAE is proposed primarily to enhance conventional imputation methods from two perspectives: the transition of training mode and the comprehensiveness of correlation extraction. The first perspective is tailored for regression-based deep learning methods, aiming to bridge the gap between the training task and imputation task. The comparative analysis of the MAMAE with other methods reveals that the two methods integrating masked autoencoding, the MAMAE and the SAITS, have advanced accuracy across all missing scenarios. Furthermore, the masking strategies of the MAMAE offers the model considerable flexibility, enabling it to achieve optimal accuracy under most missing data conditions, rather than being constrained by specific missing conditions as other common methods are. Therefore, the other methods, including the SAITS, exhibit greater degradation in accuracy when dealing with different missing data scenarios.

The second perspective of improvement aims to harness the regression-based method's ability to capture correlations at various scales, ensuring adequate information extraction for the imputation task. Specifically, two modifications are made to achieve this goal: one involves introducing turbine correlations through the feature combination, and the other involves explicitly applying the attention mechanism at different scales. As demonstrated in Section 4.3, correlations at different scales contribute differently under various missing conditions. Thus, extracting correlations across multiple scales can not only enhance accuracy but also improve the model's adaptability in diverse missing situations. Ablation experiments validate the effectiveness of the proposed modifications, with the feature combination and multiscale attention particularly beneficial for band and feature missing, respectively. To tackle the increased computational complexity resulting from the multiscale attention, the (S)W-MSA is introduced, which reduces computational complexity without sacrificing accuracy, making the modification more practical.

Future work will pursue: (i) The computational architectures of both MAMAE and SAITS are not mutually exclusive, suggesting that these two methods can potentially be integrated to further enhance accuracy. (ii) Incorporating transfer learning or meta learning into masked autoencoding to enhance the same-type and cross-type generalization capabilities of the training strategy. (iii) Leveraging the correlation extraction ability of MAMAE, an integrated architecture can be proposed for multitask O&M using data with missing values.

6. Conclusion

A multiscale attention masked autoencoder (MAMAE) was proposed

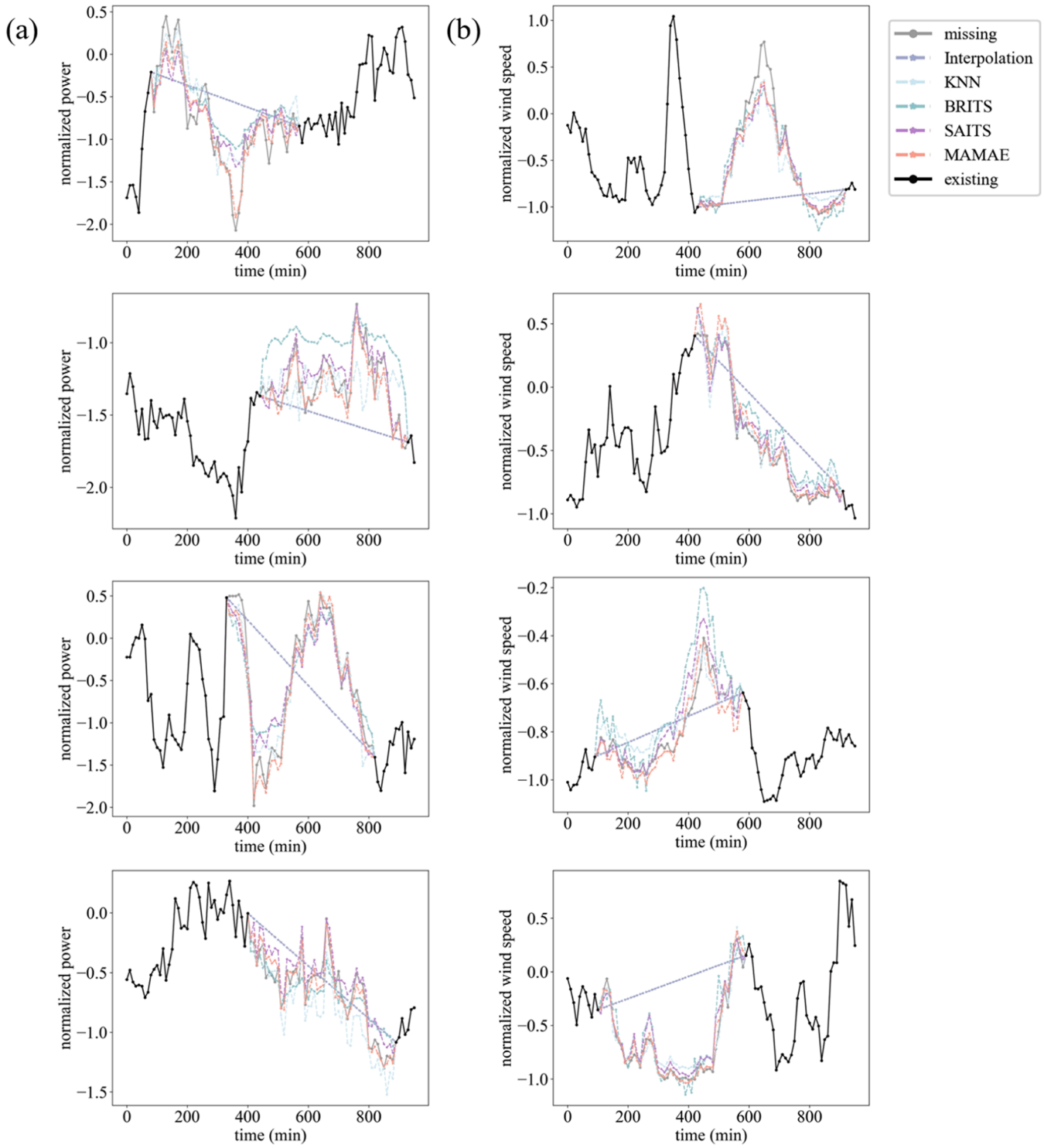

Fig. 11. Imputing results of 50 % feature missing. (a) Normalized wind speed. (b) Normalized power.

Table 5
Masked-MSE of different masking and missing ratio for random missing.

Masking ratio	Missing ratio								
	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %
10 %	0.013	0.017	0.022	0.028	0.039	0.056	0.098	0.229	0.692
20 %	0.012	0.015	0.018	0.023	0.029	0.040	0.067	0.137	0.329
30 %	**0.012**	0.014	0.017	0.021	0.025	0.033	0.048	0.091	0.314
40 %	0.013	**0.014**	0.016	0.019	0.023	0.029	0.041	0.076	0.319
50 %	0.015	0.014	**0.016**	0.018	0.022	0.026	0.034	0.055	0.201
60 %	0.016	0.015	0.016	**0.018**	**0.021**	0.025	0.031	0.045	0.128
70 %	0.019	0.018	0.018	0.019	0.021	**0.025**	**0.029**	0.039	0.074
80 %	0.039	0.029	0.025	0.024	0.025	0.026	0.030	**0.038**	0.057
90 %	0.083	0.061	0.047	0.039	0.035	0.035	0.036	0.041	**0.055**

Table 6
Masked-MSE of different masking and missing ratio for band missing.

Masking ratio	Missing ratio								
	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %
10 %	0.058	0.060	0.069	0.074	0.077	0.079	0.082	0.083	0.083
20 %	**0.046**	0.050	0.060	0.067	0.073	0.075	0.080	0.081	0.082
30 %	0.047	0.050	0.056	0.061	0.064	0.065	0.069	0.070	0.072
40 %	0.048	**0.049**	0.053	0.056	0.058	0.059	0.063	0.064	0.067
50 %	0.051	0.050	**0.053**	**0.055**	**0.057**	0.058	0.061	0.062	0.066
60 %	0.052	0.051	0.053	0.055	0.057	**0.058**	0.061	0.062	0.065
70 %	0.053	0.051	0.054	0.056	0.057	0.058	**0.061**	**0.062**	0.065
80 %	0.053	0.052	0.055	0.057	0.058	0.059	0.061	0.062	0.065
90 %	0.056	0.055	0.058	0.060	0.061	0.061	0.063	0.063	**0.064**

Table 7
Masked-MSE of different masking and missing ratio for feature missing.

Masking ratio	Missing ratio								
	10 %	20 %	30 %	40 %	50 %	60 %	70 %	80 %	90 %
10 %	0.023	0.028	0.037	0.051	0.061	0.065	0.076	0.081	0.082
20 %	**0.020**	0.020	0.025	0.036	0.049	0.054	0.065	0.073	0.076
30 %	0.020	**0.020**	0.021	0.023	0.026	0.027	0.033	0.037	0.042
40 %	0.020	0.020	0.020	0.021	0.022	0.023	0.029	0.032	0.036
50 %	0.021	0.020	**0.020**	**0.021**	0.022	0.023	0.027	0.029	0.032
60 %	0.022	0.021	0.021	0.022	0.023	0.023	0.026	0.027	0.030
70 %	0.022	0.021	0.021	0.022	0.022	0.022	0.026	0.027	0.029
80 %	0.022	0.022	0.022	0.022	**0.022**	**0.022**	**0.025**	**0.026**	0.028
90 %	0.024	0.024	0.024	0.025	0.025	0.024	0.027	0.028	**0.028**

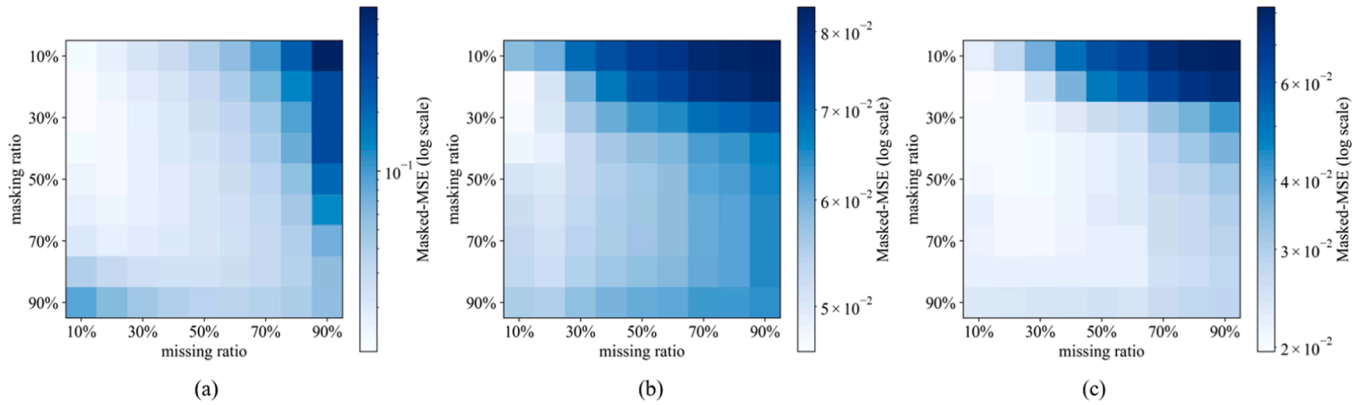

Fig. 12. The masked-MSE of the identical masking and missing type. (a) Random missing. (b) Band missing. (c) Feature missing.

Table 8
Masked-MSE of different masking and missing types for 20 % missing.

Masking type	Missing type		
	Random	Band	Feature
Random	**0.015**	0.078	0.143
Turbine	0.172	**0.050**	0.331
Feature	0.189	0.706	**0.020**

Table 9
Masked-MSE of different masking and missing types for 50 % missing.

Masking type	Missing type		
	Random	Band	Feature
Random	**0.022**	0.084	0.107
Turbine	0.200	**0.057**	0.375
Feature	0.201	0.415	**0.022**

Table 10
Masked-MSE of different masking and missing types for 80 % missing.

Masking type	Missing type		
	Random	Band	Feature
Random	**0.038**	0.113	0.077
Turbine	0.420	**0.062**	0.356
Feature	0.519	0.440	**0.026**

in this work for missing data imputation in wind turbines. The MAMAE model was trained using a self-supervised method called masked autoencoding, which artificially masks some values and tasks the model with restoring them. This approach aims to bridge the gap between the training task and the imputation task. The MAMAE model is based on the transformer architecture, and two modifications were made to account for the importance of correlations in SCADA data imputation: the feature combination and multiscale attention. The feature combination groups high-correlated turbines together, introducing turbine correlations into the data to provide additional information for imputation. The multiscale attention constructs sequences of different physical representations through token conversion and then explicitly applies the attention mechanism to the reconstructed sequences, extracting multiscale correlations for accurate and general imputation. To address the increased computational cost resulting from longer sequence lengths, the attention mechanism is implemented in shifted local windows, reducing computational complexity from quadratic to a linear relationship of the

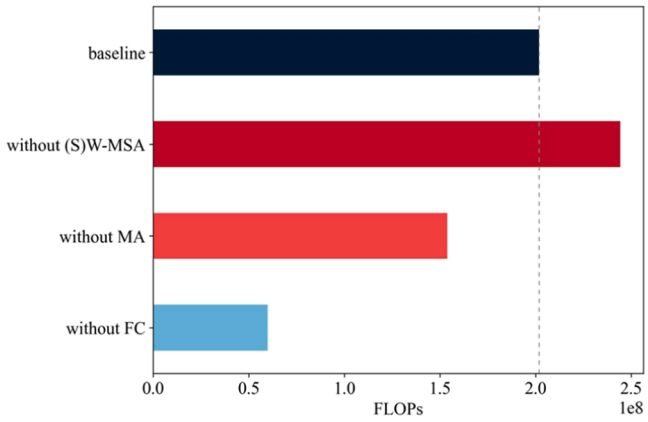

Fig. 15. The FLOPs comparison of the ablation experiment.

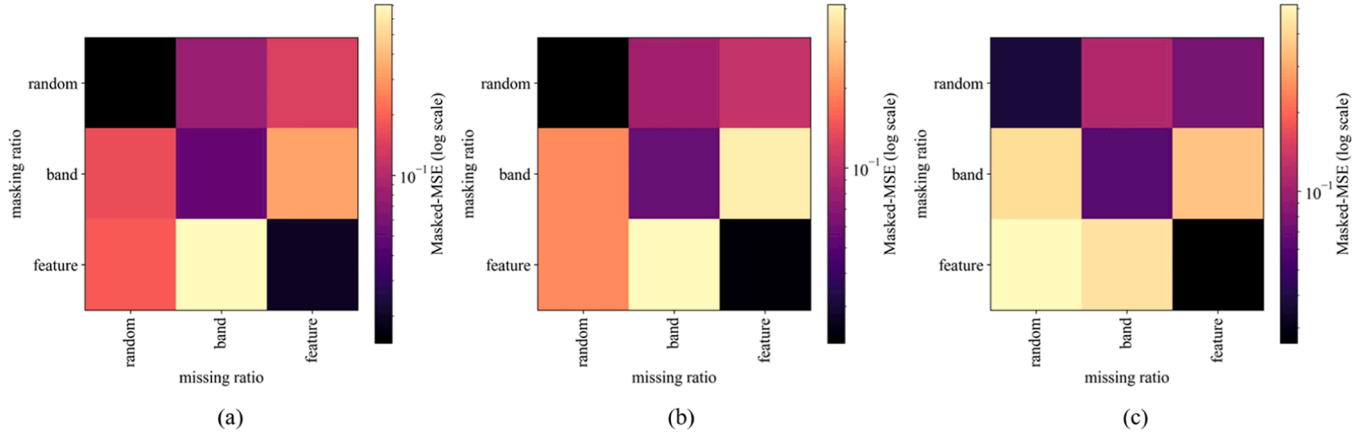

Fig. 13. The masked-MSE of the different masking types. (a) 20 % missing. (b) 50 % missing. (c) 80 % missing.

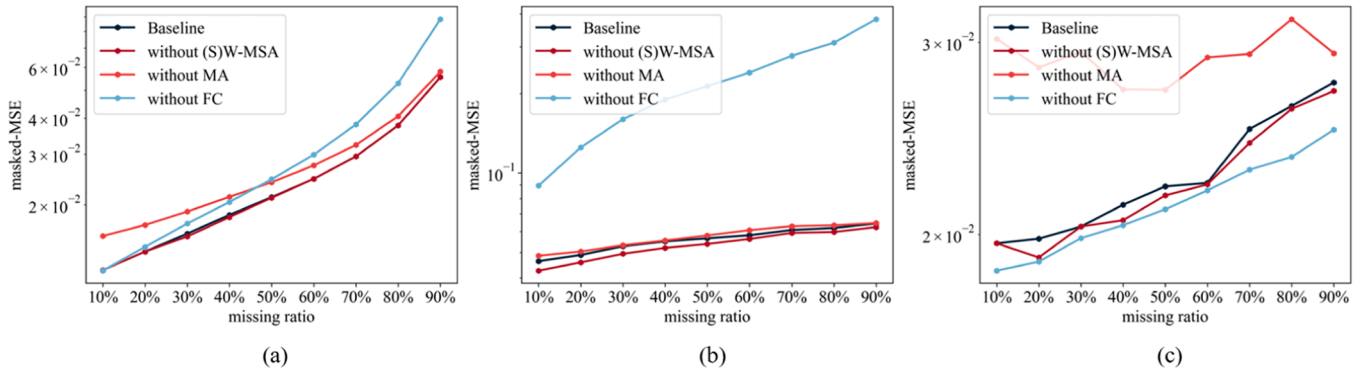

Fig. 14. The masked-MSE comparison of the ablation experiments. (a) Random missing. (b) Band missing. (c) Feature missing.

sequence length.

Numerical experiments were conducted on a real-world SCADA dataset, including accuracy assessments under various missing conditions and ablation studies of the proposed modifications. The results reveal that: (i) The MAMAE exhibits lowest MSE compared to existing imputation techniques across most missing data situations, demonstrating high accuracy, adaptability and generalization ability. (ii) The masking strategy of the masked autoencoding exhibits weak cross-type but strong same-type generalization capacity, suggesting that only a limited number of models need to be trained to cover a range of missing situations. (iii) The feature combination introduces turbine correlations into the data, substantially enhancing the imputation accuracy for band missing. (iv) The multiscale attention improves the correlation extraction capability, leading to advanced accuracy for most missing situations. The computational cost of the multiscale attention is reduced by the (S)W-MSA, with minimal loss in accuracy.

CRediT authorship contribution statement

Yuwei Fan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chenlong Feng:** Software, Methodology, Data curation. **Rui Wu:** Methodology, Investigation, Data curation. **Chao Liu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Formal analysis. **Dongxiang Jiang:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix

A.1. Instances of imputing results

A.1.1. Random missing

Figs. A-1, A-2, A-3

A.1.2. Band missing

Figs. A-4, A-5, A-6

A.1.3. Feature missing

Figs. A-7, A-8, A-9

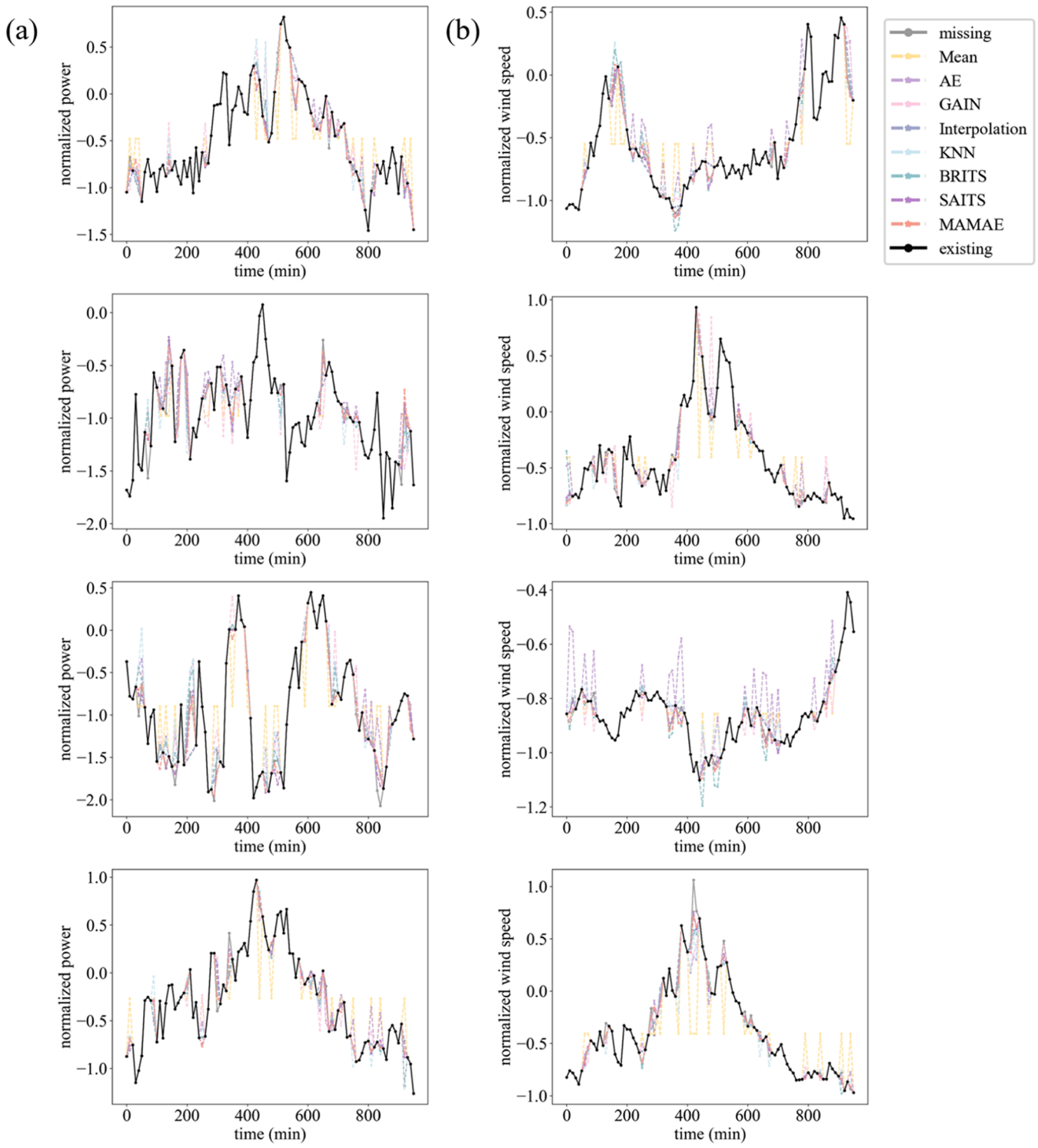

Fig. A-1. Imputing results of 20 % random missing. (a) Normalized wind speed. (b) Normalized power.

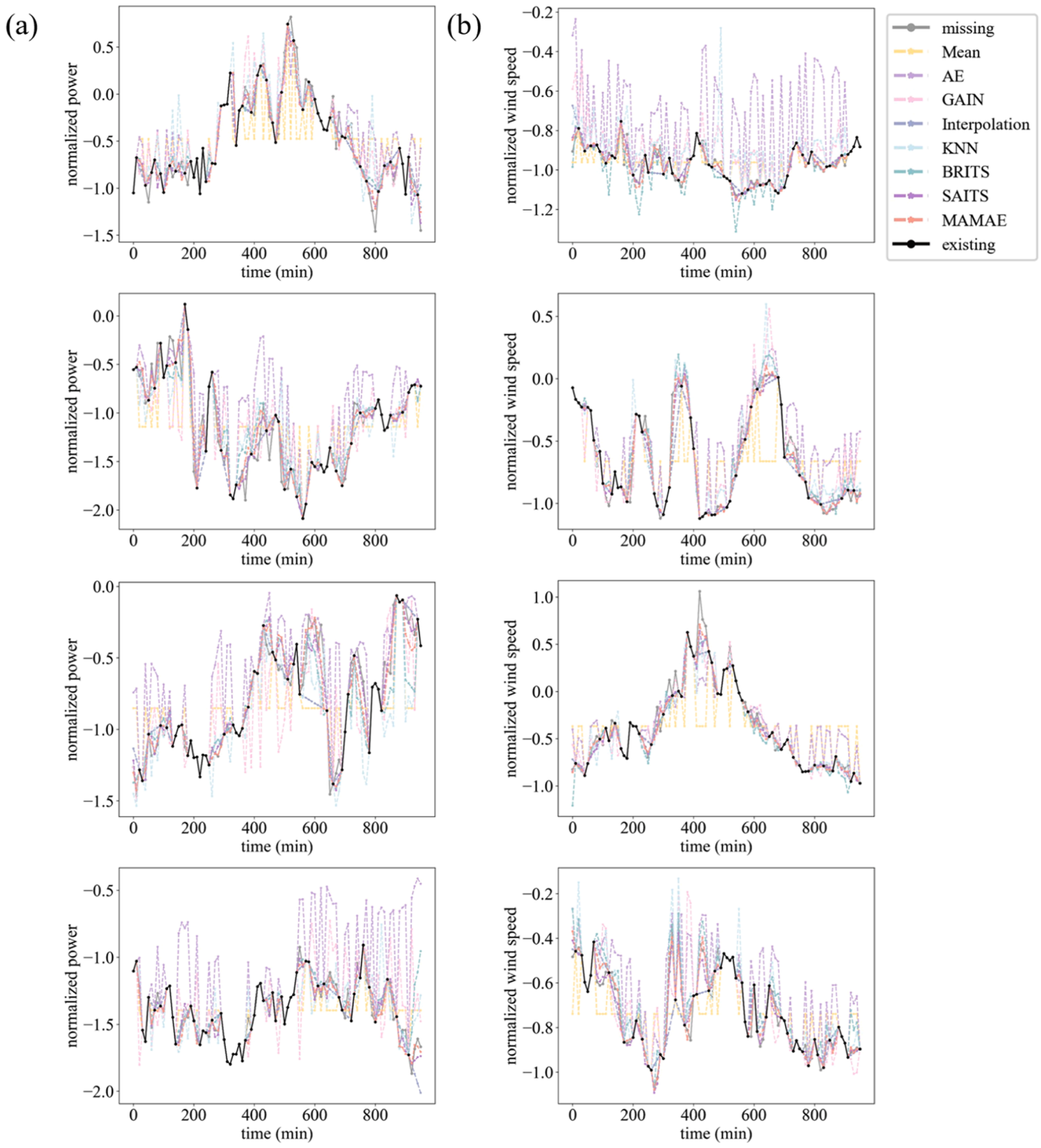

Fig. A-2. Imputing results of 50 % random missing. (a) Normalized wind speed. (b) Normalized power.

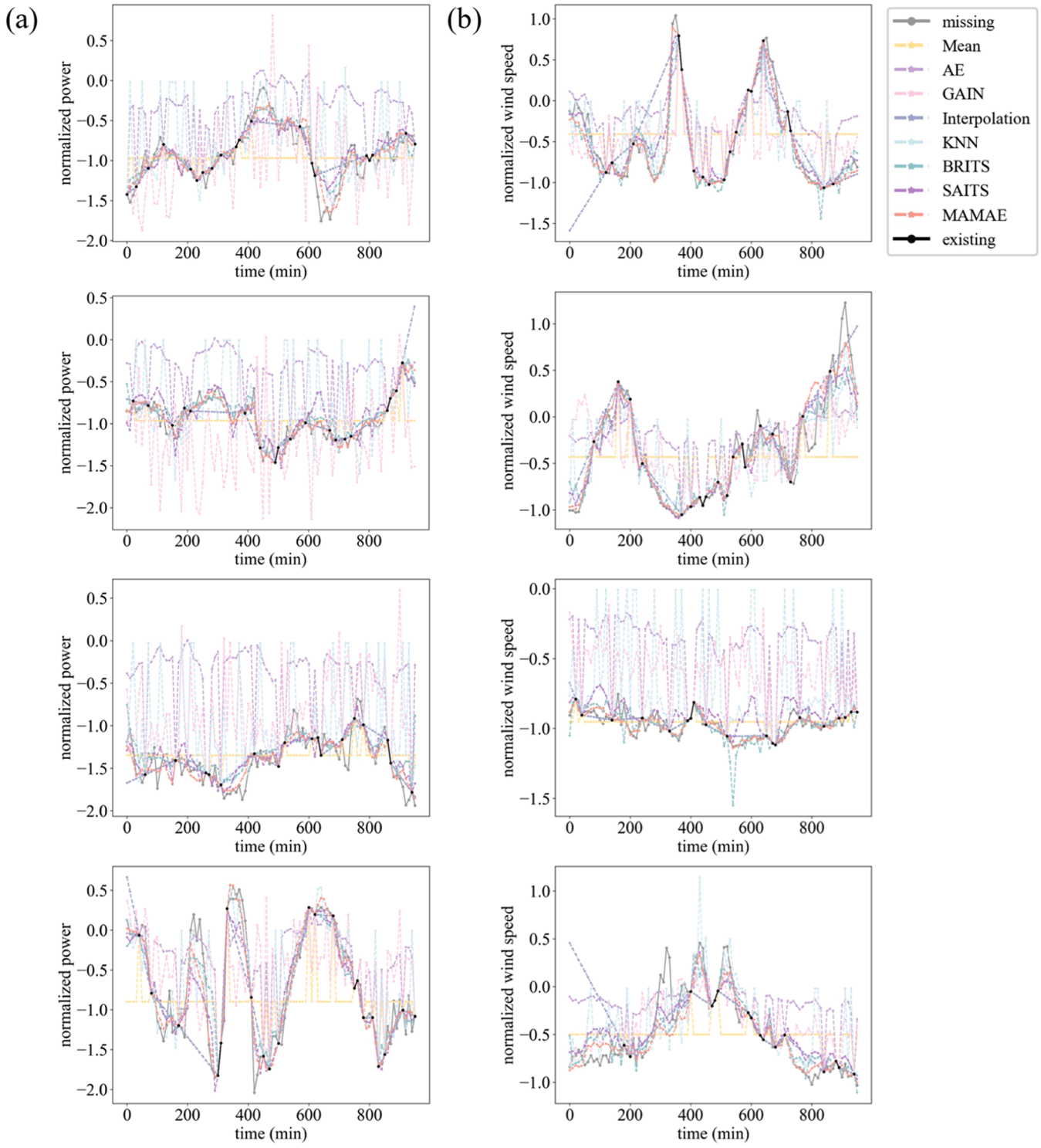

Fig. A-3. Imputing results of 80 % random missing. (a) Normalized wind speed. (b) Normalized power.

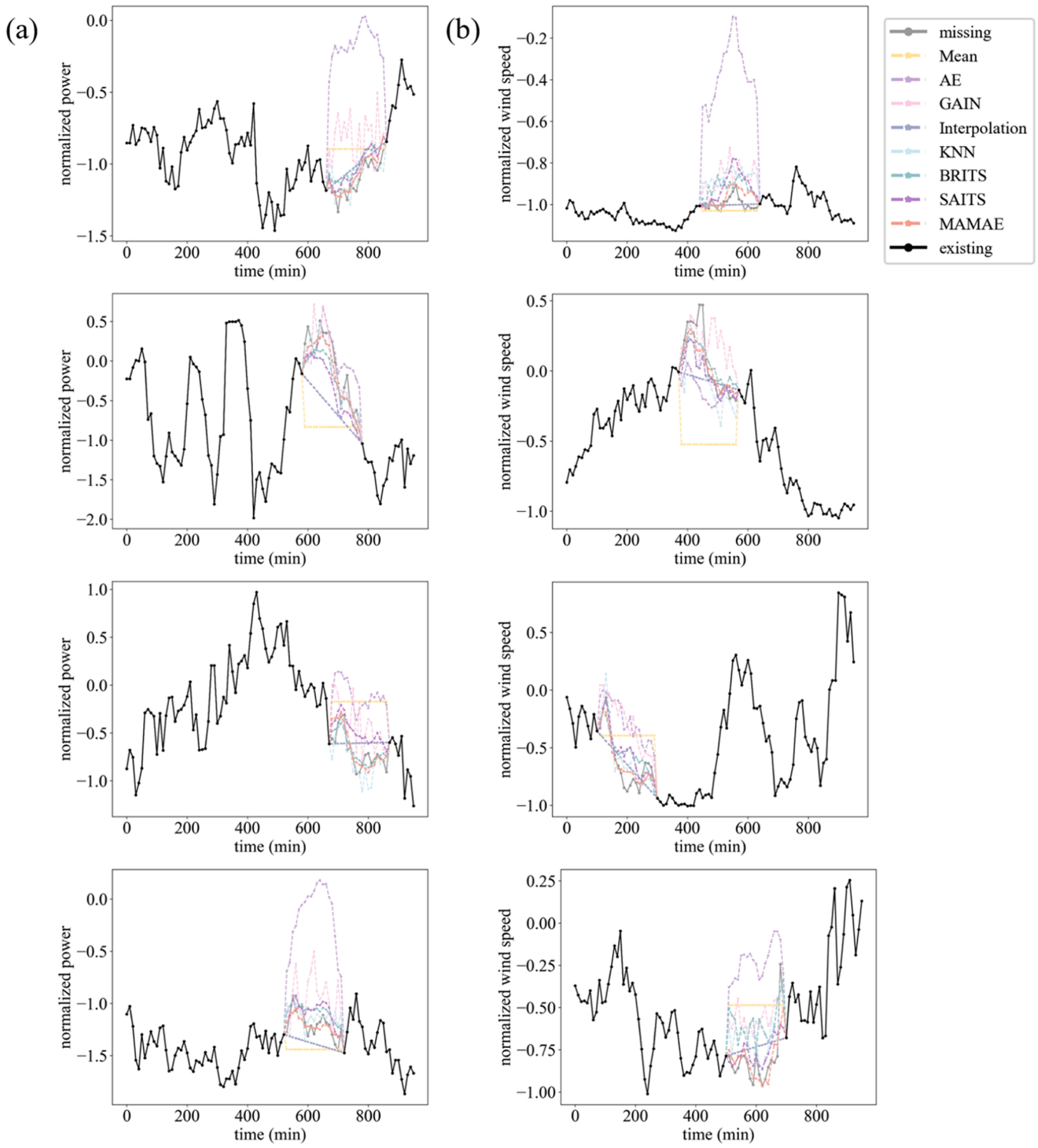

Fig. A-4. Imputing results of 20 % band missing. (a) Normalized wind speed. (b) Normalized power.

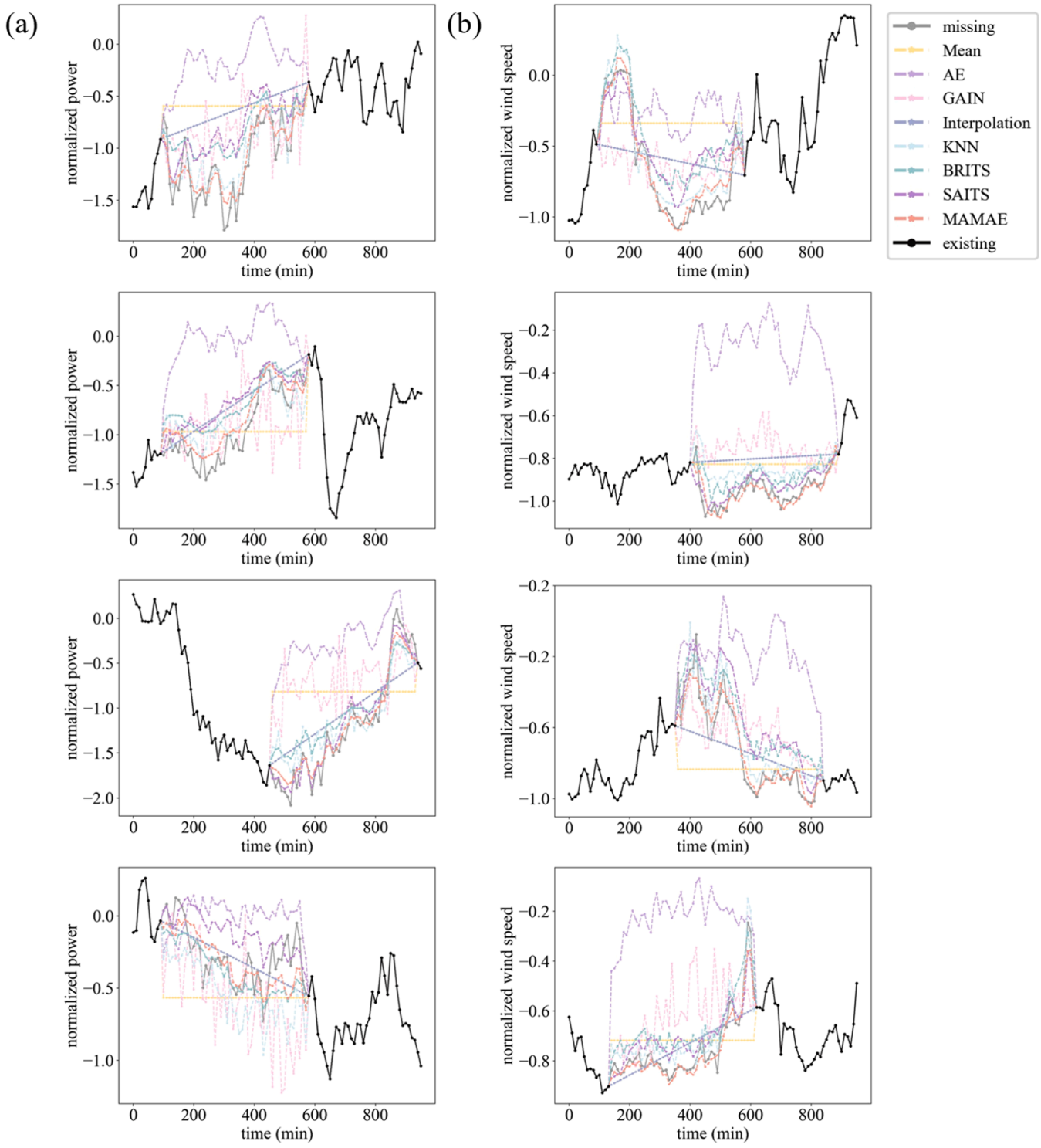

Fig. A-5. Imputing results of 50 % band missing. (a) Normalized wind speed. (b) Normalized power.

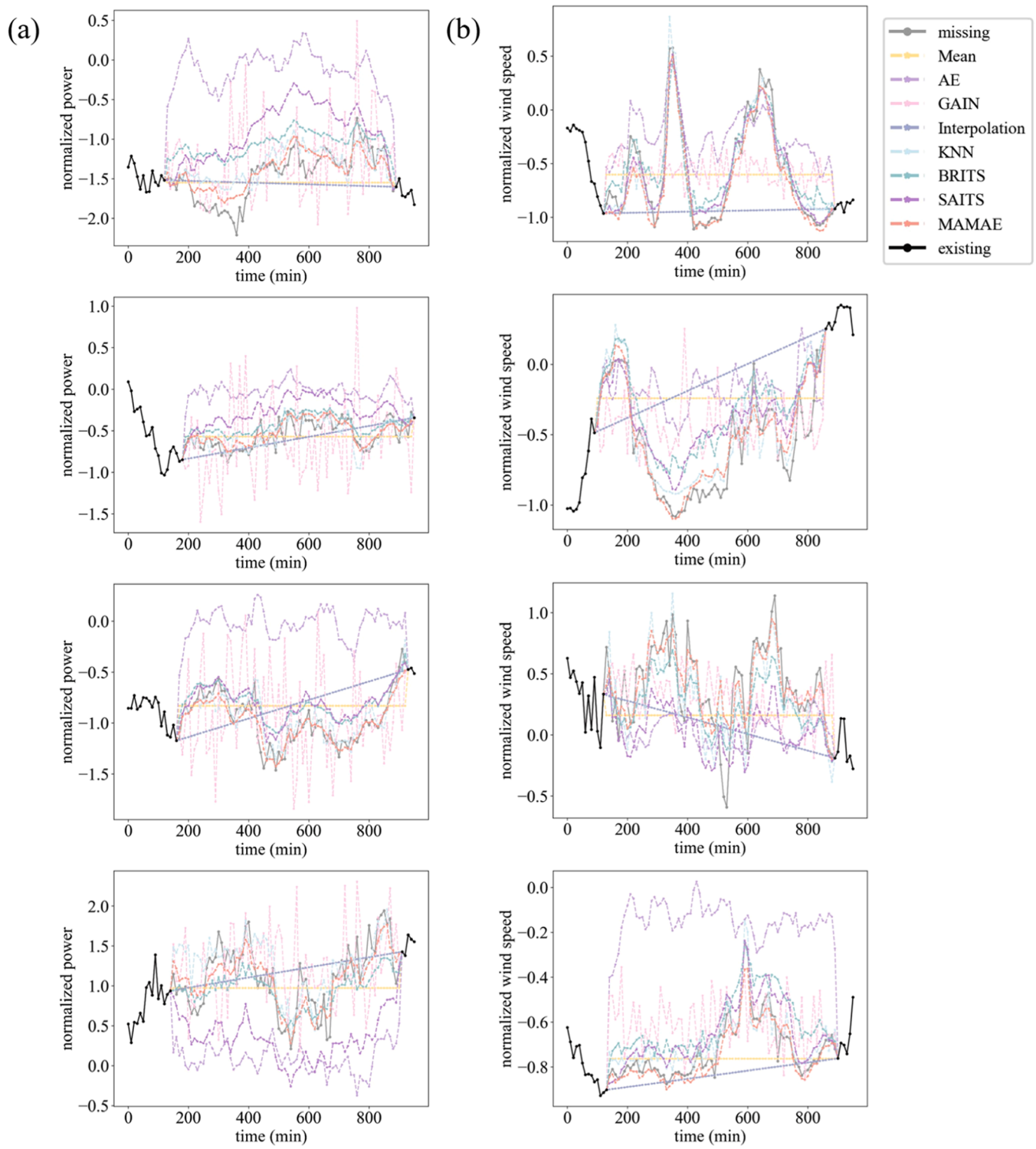

Fig. A-6. Imputing results of 80 % band missing. (a) Normalized wind speed. (b) Normalized power.

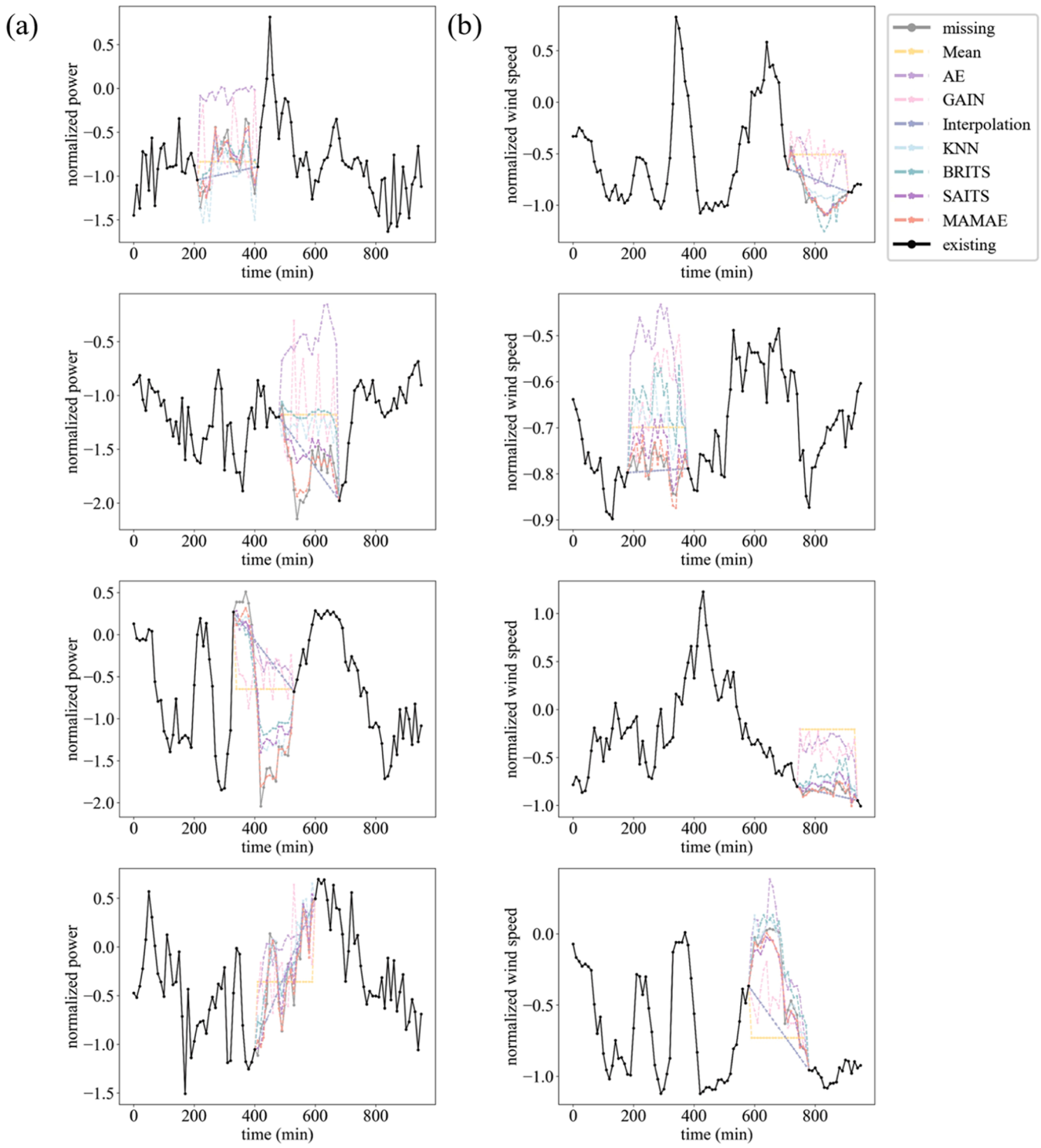

Fig. A-7. Imputing results of 20 % feature missing. (a) Normalized wind speed. (b) Normalized power.

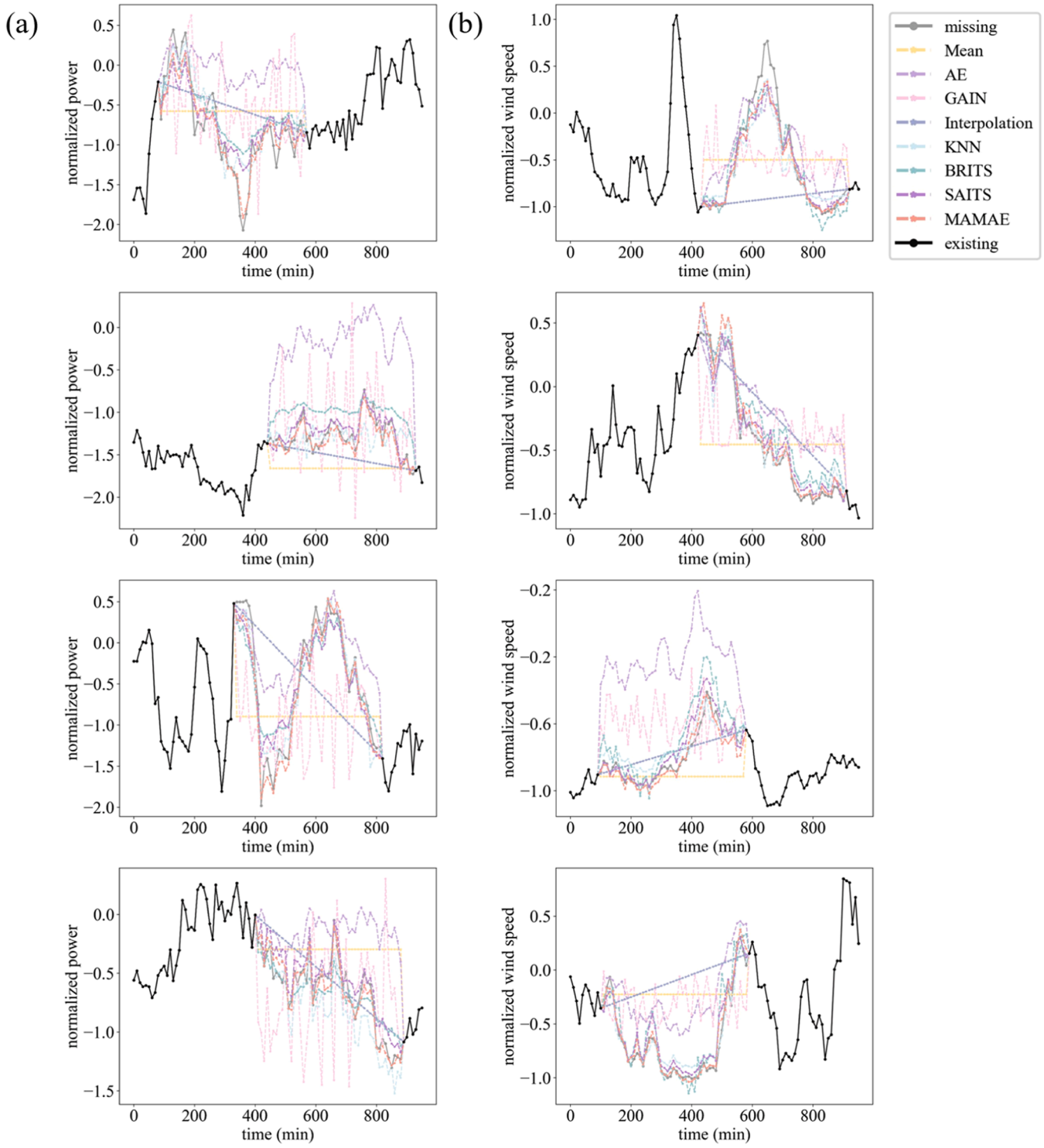

Fig. A-8. Imputing results of 50 % feature missing. (a) Normalized wind speed. (b) Normalized power.

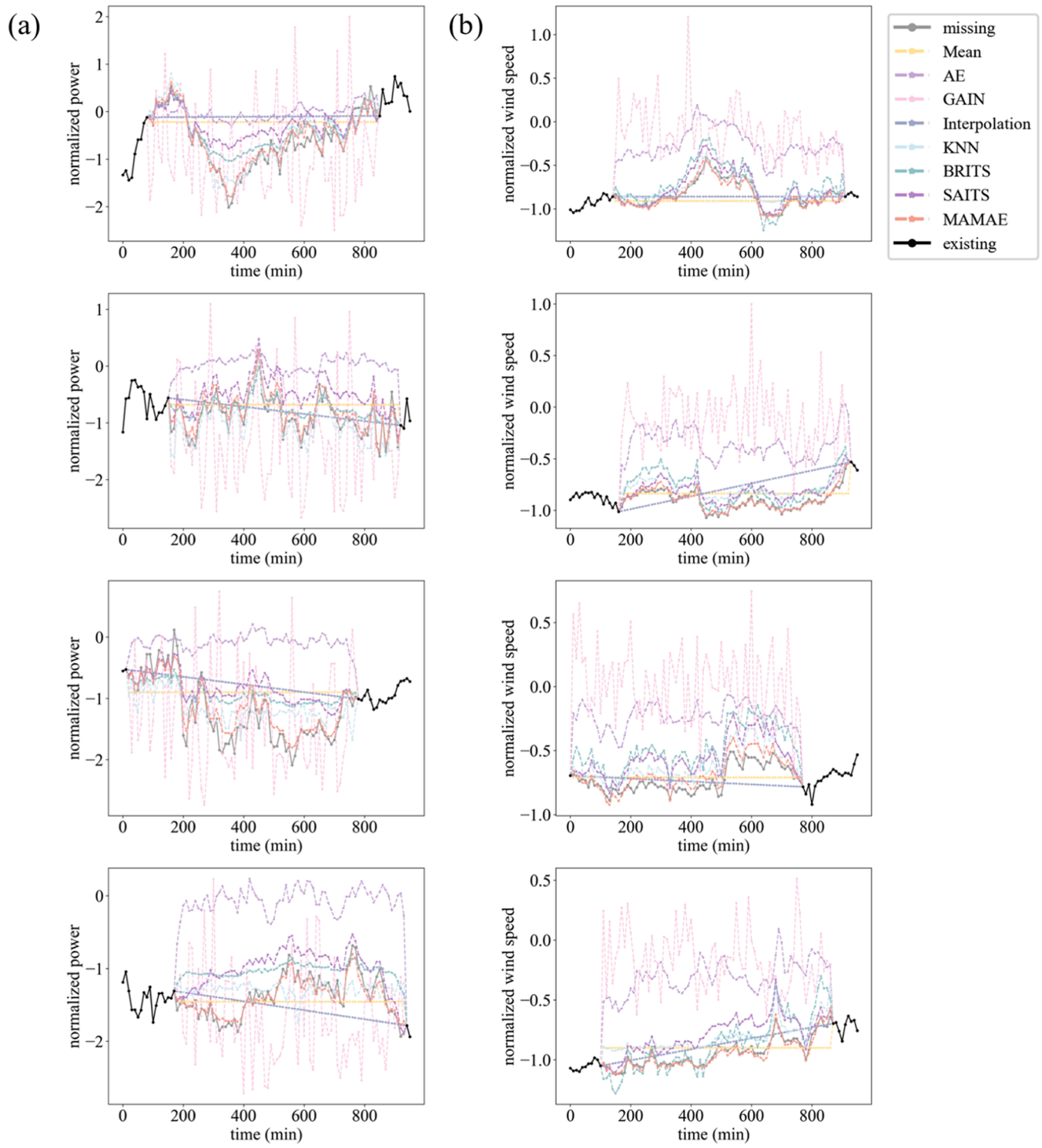

Fig. A-9. Imputing results of 80 % feature missing. (a) Normalized wind speed. (b) Normalized power.

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